

How Replicable Are Statistically Significant Findings?*

Patrick Vu[†] and Stefan Faridani[‡]

September 29, 2026

Abstract

In the empirical sciences, significance thresholds often determine whether findings are treated as evidence of an effect. This paper studies how likely findings that just meet conventional significance thresholds are to remain significant in replications of the same sample size. To answer this question, we estimate the expected replication probability conditional on a given p -value among published studies for experimental economics, psychology, and social science. We validate this measure by showing it accurately predicts actual replication outcomes, outperforming prediction markets. A finding with a p -value of 0.05 has an expected replication probability ranging from 0.10 to 0.25 across fields. Low replicability reflects small true effects in original studies rather than selective reporting. Significant findings are therefore often informative about the existence of an effect, but frequently overstate its magnitude. Finally, we develop a nonparametric estimator and apply it to economics literatures that use larger samples, finding higher but still low replication probabilities.

*We thank Johannes Abeler, Jonathan Roth, David Slichter, and seminar participants at the University of Queensland, the University of Western Australia, Georgia Institute of Technology, Boston University, Brown University, and Michigan State University for helpful comments and suggestions.

[†]Department of Economics, University of New South Wales. Email: patrick.vu@unsw.edu.au

[‡]Georgia Institute of Technology. Email: sfaridani6@gatech.edu

1. INTRODUCTION

In the empirical sciences, conventional significance thresholds are widely used to establish evidence of an effect. In some cases, a statistically significant empirical claim is repeated using the same methodology and a new sample (Open Science Collaboration, 2015; Camerer et al., 2016, 2018). A significant replication is deemed ‘successful’ and strengthens the original claim, while an insignificant replication is deemed ‘unsuccessful’ and weakens it.

Given the ubiquity of significance testing in applied research, this paper asks a simple but fundamental question: “How likely are findings that just meet conventional significance thresholds to remain significant in replications of the same sample size?” Low chances of replication are connected to several related concerns. First, many just-significant estimates may reflect favorable sampling variation around small true effects (Loken and Gelman, 2017). Second, low replication probabilities make replication ‘failures’ hard to interpret, since they could reflect either a true null or an imprecisely estimated nonzero effect (Maxwell et al., 2015). Finally, noisy and unstable estimates can reduce expected welfare from a decision-theoretic perspective, by increasing the likelihood or magnitude of decision errors (Wald, 1950).

To answer the above question, we derive the *predictive power curve* in the Andrews and Kasy (2019) framework of selective publication. Predictive power is the probability of replication for a published finding conditional on a given p -value, where replication success is defined using the standard criterion of significance at the 5% level in the same direction as the original estimate (Open Science Collaboration, 2015; Camerer et al., 2016, 2018). Within a given empirical literature, it can be interpreted as the expected replication probability of a randomly chosen published finding with that p -value.¹ We define predictive power under the original design by considering a replication with the same sample size as the original study.² Low predictive power implies that statistical significance is highly sensitive to sampling variation. Impor-

¹A p -value below 5% does not, by itself, imply a high probability of replication at the 5% level. A significant finding may reflect a large true effect, but it may also reflect a favorable sample realization from a small or null effect, in which case it is unlikely to replicate. More generally, for any p -value in $(0, 1)$, the replication probability can in principle range from 0.025 to arbitrarily close to 1.

²In practice, replication studies often use larger sample sizes than the original study, which tends to overstate replication rates relative to using the same sample size. Moreover, common power calculations that choose the replication sample size to detect the original estimate with a prespecified power yield expected replication power strictly below that target (Vu, 2024).

tantly, the framework does not require that an actual replication is conducted, or is even feasible. Predictive power is closely related to statistical power and can be applied equally to observational studies.

Predictive power has several attractive features relative to complementary measures of credibility for an empirical literature, such as the false discovery rate (FDR) and positive predictive value (PPV). First, predictive power is the probability of an observable event, so it can be empirically validated—and potentially falsified—using replication outcomes. By contrast, FDR and PPV depend on realizations of unobserved true effects and therefore cannot be directly identified from replication outcomes. Second, predictive power does not require classifying unobserved true effects as exactly null or non-null, making it less sensitive to assumptions about the distribution of effects near zero. By contrast, FDR and PPV depend critically on empirically unverifiable assumptions about whether effects are exactly zero or merely very small.

Predictive power has two noteworthy theoretical properties. First, as the conditional mean of the replication outcome, it minimizes expected squared prediction error. This provides strong theoretical grounds for using it to estimate replicability at conventional significance thresholds. Second, predictive power is invariant to selective reporting based on a finding’s own test statistic. This counterintuitive property arises because selective publication changes the distribution of observed findings *across* p -values, but not the distribution of true effects *within* a given p -value, on which predictive power is conditioned. Hence, low predictive power cannot be attributed to selective reporting.

We estimate the predictive power curve in three empirical literatures with systematic replication evidence: experimental economics ([Camerer et al., 2016](#)), psychology ([Open Science Collaboration, 2015](#)), and experimental social science ([Camerer et al., 2018](#)). Predictive power is completely determined by the distribution of true effects. We draw directly on parametric estimates of this distribution from [Vu \(2024\)](#), whose empirical applications use the ‘metastudy approach’ from [Andrews and Kasy \(2019\)](#).

The key advantage of examining literatures with systematic replication evidence is that the estimated predictive power curves can be validated against realized replication outcomes. We find that predictive power forecasts actual replication outcomes with remarkable accuracy: we cannot reject perfect calibration, and it achieves lower average squared prediction error than prediction markets, despite using far less infor-

mation. High predictive accuracy provides strong empirical justification for using the measure to evaluate replicability, and it also suggests a simple, low-cost alternative to prediction markets for forecasting replication outcomes.

Having validated our measure, we next turn to the paper’s main question: how replicable are findings that narrowly satisfy conventional significance thresholds when replicated using the original sample size? In all three applications, and across all standard conventional significance thresholds, the estimated predictive power curves point to the same conclusion: replication probabilities are generally quite low. For example, in experimental economics—which fares better than psychology and social science—a finding with a p -value of 0.05 has a one-in-four chance of being successfully replicated. This does not imply that three-quarters of just-significant findings are false positives. Nor do unsuccessful replications necessarily imply errors in original studies. Instead, low replicability can be explained almost entirely by the winner’s curse ([Andrews et al., 2024](#)). Specifically, just-significant estimates often reflect small underlying true effects with ‘favorable’ sampling variation that makes them significant. Consistent with this mechanism, we estimate that 84% of experimental economics findings with $p = 0.05$ overstate the magnitude of their true effects.

The predictive power curve also allows us to evaluate replicability for other p -values. At $p = 0.01$, the expected replication probability at the 5% level increases, although remains relatively low, at 0.40. The share of findings estimated to overstate the magnitude of their true effects remains high, at 83%. Notably, moving from “marginal significance” at the 10% level to statistical significance at the 5% level only increases expected replication probabilities from 0.19 to 0.25, implying little change in replicability. More generally, achieving high replication probabilities requires far stronger statistical evidence than demanded by conventional thresholds. For example, achieving a replication probability of at least 0.8 in experimental economics requires $p < 0.00006$, a standard met by 16.7% of findings in the sample. We interpret this as a diagnostic benchmark rather than a recommendation to impose stricter thresholds.

To clarify what these results imply for evidence against the null, Subsection 4.2 examines how a successful or unsuccessful replication updates the probability that the true effect is ‘close to zero’, beginning with an original finding with $p = 0.05$.³

³Our estimated distribution of true effects is continuous and therefore assigns zero probability to an effect of exactly zero. We instead consider a near-null effect, defined as one sufficiently small to have less than a 5% probability of significance in a same-sized replication.

For experimental economics, we find that a successful replication can substantially strengthen evidence against a near-null effect, reducing its probability from 0.16 to 0.02. By contrast, an unsuccessful replication provides only limited evidence in its favor, increasing the probability from 0.16 to 0.21. These results highlight two points. First, most just-significant findings are relatively unlikely to reflect near-null effects, even though their estimated magnitudes are frequently overstated. Second, replication outcomes are highly asymmetric in their informational content: a successful replication provides strong evidence against a near-null effect, whereas an unsuccessful replication provides comparatively little evidence in its favor. This asymmetry cautions against interpreting low replication rates as evidence that a large share of original findings are false.

To broaden the applicability of our framework, we develop a nonparametric estimator of predictive power based on deconvolution under selective reporting. Its principal advantages are that it imposes no parametric assumptions on the distribution of true effects and can also be used in settings where each study reports multiple correlated p -values. Its main drawback is that it cannot be applied to the replication studies due to the small number of observations. Instead, we apply it to a large dataset of experimental and quasi-experimental findings in economics (Brodeur et al., 2020). For randomized control trials, the chance of replicating a finding with $p = 0.05$ is 0.34. Quasi-experimental studies using observational data have notably higher replication probabilities at $p = 0.05$, ranging from 0.43 to 0.50, although replicability remains relatively low in absolute terms. Across both parametric and nonparametric applications, differences in predictive power can largely be explained by differences in sample size.

Overall, the results in this paper suggest that a significant study may be informative about the existence of an effect but much less reliable about its magnitude or reproducibility. These findings support the longstanding view that robust scientific conclusions require cumulative evidence across multiple studies (Popper, 1934), and reinforce more recent calls to strengthen incentives for conducting and publishing replications (Nosek et al., 2012; Coffman and Niederle, 2015; Brodeur et al., 2023).

Related Literature. The paper contributes to two strands of the metascience literature. First, it contributes to research on replications and statistical power (Cohen, 1988; Ioannidis, 2005; Loken and Gelman, 2017; Ioannidis et al., 2017; Andrews and Kasy, 2019; DellaVigna and Linos, 2022). It is most closely related to two re-

cent papers. First, [van Zwet and Goodman \(2022\)](#) estimates predictive power in a parametric model without publication bias for medical research and finds low replicability. Second, [Lang \(2025\)](#) estimates 65% of narrowly rejected null hypotheses in economics are likely to be false rejections. This paper builds on the literature in several respects. First, to our knowledge, it is the first study to validate predictive power against actual replication outcomes. Second, it models flexible forms of publication bias, for which there is substantial empirical evidence across scientific fields ([Card and Krueger, 1995](#); [Franco et al., 2014](#); [Andrews and Kasy, 2019](#)). Finally, it develops a nonparametric estimator of predictive power under weak assumptions.

Second, the paper contributes to the literature on predicting replications ([Dreber et al., 2015](#); [Altmejd et al., 2019](#); [DellaVigna et al., 2019, 2020](#); [Gordon et al., 2021](#)). Existing approaches use surveys, prediction markets, and machine learning methods to forecast replication outcomes. Our simple model-based forecast uses p -values from original studies yet nevertheless outperforms prediction markets. Thus, it offers a simple, low-cost alternative to more resource-intensive forecasting methods.

The remainder of the article is organized as follows. Section 2 derives predictive power. Section 3 discusses estimation and validation. Section 4 presents the parametric results and Section 5 the nonparametric results. Section 6 makes recommendations for researchers and replicators. Section 7 concludes.

2. THEORY

This section introduces the framework, derives the predictive power curve, and discusses its properties. Proofs are in Appendix A.

2.1 Setup

Consider an empirical literature composed of research findings indexed by i . For instance, the literature of interest could be based on field (e.g. labor economics, electoral politics, epidemiology) or methodology (e.g. experimental, quasi-experimental). Research findings are characterized by an (unobserved) true effect, an estimate of the true effect, and a standard error, $(\beta_i, \hat{\beta}_i, \sigma_i)$.⁴ We focus on normalized studies,

⁴Following [Andrews and Kasy \(2019\)](#) and [Elliott et al. \(2022\)](#), we assume for simplicity that the researcher observes the true standard error. Under standard regularity conditions, estimation error in the standard error is typically much smaller than sampling variation in the effect estimate.

characterized by $(z_i, \hat{z}_i)$, where $z \equiv \beta/\sigma$, $\hat{z} \equiv \hat{\beta}/\sigma$. For expositional simplicity, we refer to z as the *true effect* and $\hat{z}$ as the *estimated effect*.

The data-generating process follows the selective publication model of [Andrews and Kasy \(2019\)](#) presented in normalized units:

1. **Draw true effect:** Draw z from the distribution of true effects: $z \sim \pi$.
2. **Estimate the effect:** Draw an estimate $\hat{z}$ from a normal distribution centered at the true effect z : $\hat{z} \mid z \sim N(z, 1)$.
3. **Publication selection:** Selective publication is modeled by a function $s(\cdot)$, which gives the probability of publication for an estimate $\hat{z}$. Let D be a Bernoulli random variable equal to 1 if a finding is published and 0 otherwise. Then $\mathbb{P}(D = 1 \mid \hat{z}) = s(\hat{z})$.

We observe $|\hat{z}|$ for published findings, $D = 1$. Focusing on the absolute value reflects common empirical settings in which only the magnitude of the test statistic – or, equivalently, the two-sided p -value – is reported.

To illustrate the model, consider the quasi-experimental labor economics literature. Papers in this literature address different questions, such as the impact of the minimum wage on employment or the effect of a job-training program on earnings. Each question has an associated true effect z , drawn from the distribution $\pi(z)$.⁵ For the job-training example, z represents the program’s true average treatment effect, normalized by the study’s standard error. In the second step, a researcher applies an empirical method – such as a difference-in-differences design – to obtain the estimate $\hat{z}$. Under mild regularity conditions, the resulting estimator is approximately normally distributed with a consistently estimable variance, a standard assumption used in empirical practice. In the final step, the estimate $\hat{z}$ is published with probability $s(\hat{z})$. This captures selective publication; for instance, statistically significant findings may be more likely to be published than null results.

2.2 Replication Under Original Design

Our main objective is to estimate the probability that a finding with a given p -value successfully replicates. Formally, consider a replication of an original finding $(z, \hat{z})$.

⁵While $\pi(z)$ can be interpreted as a ‘prior’ in a Bayesian hierarchical model, it is not subjective. Instead, it should be interpreted as the objective distribution of true effects across studies in the empirical literature of interest.

To evaluate replicability under the original design, we consider a replication that uses the same sample size as the original study. Thus, the replication estimate satisfies $\hat{z}_r \mid z \sim N(z, 1)$ and is conditionally independent of $\hat{z}$ given z .

Replication success is defined using the standard criterion from large-scale replication studies: statistical significance at the two-sided 5% level with the same sign as the original estimate (Open Science Collaboration, 2015; Camerer et al., 2016, 2018). Specifically, let the replication indicator be $R = \mathbb{1}\{|\hat{z}_r| \geq 1.96 \text{ and } \text{sign}(\hat{z}_r) = \text{sign}(\hat{z})\}$, where 1.96 is the critical value for statistical significance at the two-sided 5% level. This measure of replication success is closely related to statistical power.⁶ As with statistical power, it is well-defined for observational studies for which a direct replication may be infeasible.

We focus on replications using the same sample size as original studies. In practice, replication studies use various sample-size rules, most of which lead to larger samples and therefore higher replication rates.⁷ Our benchmark instead evaluates replicability under the original research design, which provides a common basis for comparison across literatures that is independent of replicators' sample-size choices.

2.3 Predictive Power Curve

Define *predictive power* as the probability of a successful replication conditional on the observed two-sided p -value and publication: $r(p) \equiv \Pr(R = 1 \mid p, D = 1)$. Tracing $r(p)$ over $p \in (0, 1)$ yields the *predictive power curve*. We express predictive power in terms of the two-sided p -value because statistical significance is typically reported using two-sided thresholds such as $p < 0.05$ or $p < 0.01$. This is without loss of generality because the two-sided p -value maps one-to-one to $|\hat{z}|$.

It is useful to distinguish predictive power from other common definitions of power. *Planned power* is the probability of obtaining a statistically significant result under a

⁶Conventional one-sided power is the probability that the replication estimate is significant with the same sign as the true effect, whereas our measure requires it that be the same sign as the original estimate, in line with the conventional approach used in replication studies. The two coincide unless the original estimate is statistically significant with the wrong sign, and are therefore similar when such sign errors are rare.

⁷Appendix D estimates that the rules for setting replication sample sizes in the experimental economics (Camerer et al., 2016), psychology (Open Science Collaboration, 2015), and experimental social science (Camerer et al., 2018) replication projects raise estimated replication rates by 30–36% relative to the same-sample-size benchmark. More generally, replication rates depend mechanically on the chosen sample-size: for any non-zero effect, statistical power approaches one as sample size increases without bound and approaches the size of the test as sample size approaches zero.

prespecified effect size. *Actual power* is this probability under the (unobserved) true effect. Finally, *post hoc power* substitutes the original estimate for the true effect. As a one-to-one transformation of the p -value, post hoc power provides no additional information about the evidence contained in the original estimate (Hoenig and Heisey, 2001).

The following Lemma gives a convenient representation of predictive power:

Lemma 1 (Predictive Power). *Let $s(\cdot)$ be symmetric about zero and strictly positive. Then for every $p \in (0, 1)$, predictive power is equal to*

$$r(p) = \frac{\int_{\mathbb{R}} [1 - \Phi(1.96 - z)] \varphi(cv(p) - z) \tilde{\pi}(z) dz}{\int_{\mathbb{R}} \varphi(cv(p) - z) \tilde{\pi}(z) dz} \quad (1)$$

where $cv(p) \equiv \Phi^{-1}(1 - p/2)$ and $\tilde{\pi}(z) \equiv \frac{\pi(z) + \pi(-z)}{2}$ is the symmetrized distribution of normalized true effects.

Predictive power in (1) can be interpreted as the probability that a randomly chosen published finding with a given p -value is statistically significant at the 5% level in the same direction as the original estimate in a replication using the same sample size as the original study.

The assumptions that $s(\cdot)$ is symmetric about zero and strictly positive are relatively mild. Symmetry requires that publication depend on the magnitude rather than the sign of the test statistic. This is natural in the empirical literatures we examine, where different estimates corresponds to different research questions. As a result, there is little reason for positive estimates to be more publishable than negative ones. Strict positivity rules out regions of the test-statistic distribution in which publication is impossible. In practice, this restriction is satisfied in all the datasets we examine, since they contain published findings throughout the p -value distribution. We express predictive power in terms of the symmetrized distribution because $\pi(\cdot)$ is not identified from absolute test statistics—only its symmetrization $\tilde{\pi}(\cdot)$ is identified (Theorem B.1). Nonetheless, the representation in Lemma 1 is exact since predictive power is invariant to replacing $\pi(\cdot)$ with $\tilde{\pi}(\cdot)$.

Predictive power has three noteworthy features. First, since R is binary, $r(p)$ is the conditional expectation of replication success given the original p -value, and therefore minimizes expected squared prediction error. This provides a theoretical justification for focusing on predictive power to evaluate expected replicability at conventional

significance thresholds.

Second, predictive power does not depend on the publication bias function $s(\cdot)$. It is therefore invariant to arbitrary selective reporting based on the magnitude of a finding’s own test statistic. The reason for this counterintuitive property is that selective publication changes the relative frequency of different p -values, but not the distribution of power conditional on a given p -value. In other words, publication bias treats all findings with a given p -value identically and therefore leaves the expected replication probability conditional on that p -value unchanged. This invariance has substantive implications for interpreting $r(p)$. Specifically, expected replicability at any fixed p -value is identical whether such selection is highly prevalent or entirely absent; hence, low predictive power cannot be attributed to publication bias.

Third, predictive power is completely characterized by the symmetrized distribution of normalized true effects, $\tilde{\pi}(z)$. Consequently, differences in predictive power across literatures arise entirely from differences in $\tilde{\pi}(z)$. For instance, in a literature containing mostly large true effects, predictive power will be relatively high for a given p -value. By contrast, if $\tilde{\pi}(z)$ places substantial mass near zero, predictive power will be relatively low for the same p -value. The central empirical task is therefore to estimate the symmetrized distribution of normalized true effects.⁸ In the next section, we present results based on parametric estimates of $\tilde{\pi}(z)$, and in Section 5 we develop a novel nonparametric estimator.

3. ESTIMATION AND VALIDATION

This section estimates the predictive power curve for three empirical literatures and validates its performance against observed replication outcomes.

3.1 Data

We examine three large-scale replication projects that systematically replicated published studies from leading journals. First, [Camerer et al. \(2016\)](#) replicate experimental economics results from all 18 between-subjects laboratory experiments published in *American Economic Review* and *Quarterly Journal of Economics* between 2011

⁸Following [Andrews and Kasy \(2019\)](#), we treat $\pi(z)$ throughout as a density. This simplifies the exposition and the proofs for the nonparametric results. The analysis can be extended to discrete or mixed distributions of true effects without changing the nonparametric estimator, similarly to [Faridani \(2026\)](#).

and 2014. Second, [Open Science Collaboration \(2015\)](#) replicate results from 100 psychology findings in 2008 from *Psychological Science*, *Journal of Personality and Social Psychology*, and *Journal of Experimental Psychology: Learning, Memory, and Cognition*. Following [Andrews and Kasy \(2019\)](#), we consider a subsample of 73 studies with test statistics that are well-approximated by z -statistics. Finally, [Camerer et al. \(2018\)](#) replicate 21 experimental studies in the social sciences published between 2010 and 2015 in *Science* and *Nature*.

3.2 Parametric Estimation

Predictive power in equation (1) is fully determined by the symmetrized distribution of true effects, $\tilde{\pi}(z)$. While publication bias does not enter the expression for $r(p)$, it must still be accounted for when estimating $\tilde{\pi}(z)$ from published findings. Ignoring it would generally yield inconsistent estimates of $\tilde{\pi}(z)$, and hence of $r(p)$. To estimate $\tilde{\pi}(z)$, we use the metastudy approach of [Andrews and Kasy \(2019\)](#), with parameter estimates taken from Table 1 in [Vu \(2024\)](#). Estimates from [Vu \(2024\)](#) are used because [Andrews and Kasy \(2019\)](#) estimate the distribution of true effects, β , but not of standard errors, σ ; hence, their estimates do not recover the distribution of z required to calculate predictive power.

The metastudy approach corrects for publication bias and assumes that true effects and standard errors are independent, each following a gamma distribution. It treats findings within each empirical literature as i.i.d. draws from the model described in Subsection 2.1. It normalizes true effects to have positive support. We symmetrize the estimated distribution around zero before calculating predictive power, following Lemma 1. This is a normalization rather than a substantive restriction.⁹

A key advantage of the metastudy approach is that it only uses data from original studies in estimation, allowing us to validate predictive power out of sample against observed replication outcomes. Its main limitation is the assumption that true effects and standard errors are independent. In Section 5, we develop a nonparametric estimator that relaxes both this independence assumption and the parametric assumptions on $\tilde{\pi}(z)$.

For robustness, Figure C1 repeats the analysis using the systematic-replication ap-

⁹Equivalently, we can retain the nonnegative distribution of true effects and calculate the probability that the replication estimate is significant with the same sign as the original estimate. This yields identical results.

proach of [Andrews and Kasy \(2019\)](#), which combines original and replication studies and relies on weaker identifying assumptions. In particular, it does not require independence between β and σ . Since replication data enter the estimation, this approach cannot be validated out of sample against those same replication outcomes. The resulting predictive power curves are quantitatively similar to those obtained using the metastudy approach, providing further support for the metastudy estimates.

3.3 Validation

Does predictive power accurately predict whether original studies actually replicate? Answering this question is central to establishing the empirical validity of the measure. If predictive power predicts observed replication outcomes, then it validates the measure and justifies its use in evaluating replicability. Conversely, if it fails to predict observed outcomes, any conclusions drawn from it lack empirical support and should be interpreted with caution.

Unlike the same-sample-size benchmark used in our main analysis, the validation exercise uses the sample-size rules actually implemented in the replication studies. We therefore construct predictive power curves for each field using a modified version of equation (1) that replaces original-study standard errors with replication standard errors. This is feasible because replication sample sizes in each application were set using prespecified rules.¹⁰ See Lemma A.1 in the Appendix for details.

To assess predictive accuracy, we estimate the following linear probability model:

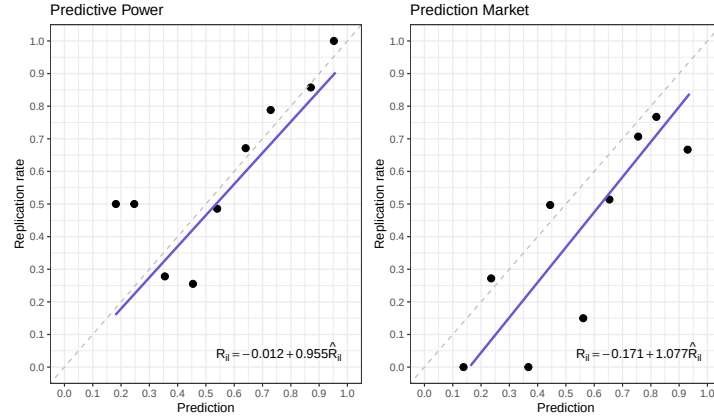
$$R_{il} = \alpha + \gamma \hat{R}_{il} + \epsilon_{il} \quad (2)$$

where R_{il} is an indicator equal to one if study i in empirical literature l (i.e. economics, psychology, or social science) successfully replicates with significance and the same sign as the original estimate, and zero otherwise; and $\hat{R}_{il}$ is its predicted replication probability. Estimation assigns equal weight to each literature.

We compare predictive power based on the metastudy method with prediction-market forecasts compiled by [Gordon et al. \(2021\)](#). Before the replications were conducted, academics traded contracts whose final market prices represent aggregate predicted replication probabilities. Prediction markets were run for a subset of repli-

¹⁰Specifically, in equation (1), we replace $1 - \Phi(1.96 - z)$ with $1 - \Phi(1.96 - z_r)$, where $z_r = \beta/\sigma_r$ and σ_r is determined by the replicators' sample-size rule.

FIGURE 1. Replication Prediction Accuracy



Notes: The figure presents binscatter plots corresponding to equation (2) for predictions from predictive power and prediction markets. Solid lines show fitted values from estimating equation (2), and dashed lines show the 45-degree benchmark. Perfect calibration is assessed using a joint test of $\alpha = 0$ and $\gamma = 1$. The corresponding p -values are 0.762 for predictive power and 0.084 for prediction markets. Prediction-market forecasts are from [Gordon et al. \(2021\)](#).

cated findings; and the validation exercise uses the sample of 70 studies for which market predictions are available.

Figure 1 presents the results graphically using binscatter plots with fitted lines. Visually, both predictors are highly predictive of realized replication outcomes. Remarkably, the fitted line for predictive power lies almost exactly on the 45-degree benchmark of perfect calibration, where predicted replication probabilities equal realized replication rates. By contrast, the fitted prediction-market line lies below the benchmark, indicating that prediction markets tend to be overly optimistic about replication success. Formally, a joint test of perfect calibration ($\alpha = 0$, $\gamma = 1$) does not reject the null for predictive power ($p = 0.762$), but marginally rejects it for prediction markets at the 10% level ($p = 0.084$).

Table 1 complements the calibration regressions by comparing predictors using average squared prediction error, also known as the Brier score.¹¹ Calibration in Figure 1 assesses whether predicted probabilities match realized replication rates on average, whereas the Brier score also rewards forecasts that distinguish between studies with different replication prospects. For example, assigning every study a probability of

¹¹The Brier score is strictly proper: a forecaster minimizes expected loss by reporting their true subjective probability. This makes it suitable for comparing probability forecasts because it rewards honest, well-calibrated probabilities rather than strategic overstatement or understatement.

TABLE 1 – Replication Forecasting Error

	Predictive Power	Prediction Market
Economics	0.188	0.226
Psychology	0.243	0.243
Social Science	0.217	0.190
All	0.216	0.220

Notes: This table reports Brier scores for predictive power and prediction-market forecasts. For finding i , the Brier score is $(R_i - \hat{R}_i)^2$, averaged across findings. Lower scores indicate greater predictive accuracy. The final row weights each replication project equally.

0.30 is perfectly calibrated if 30% replicate, but provides no information about which studies are more likely to replicate. The Brier score addresses this potential shortcoming by penalizing prediction errors for each study, thereby rewarding forecasts that are both accurate on average and informative at the individual-study level.

In the pooled sample, predictive power slightly outperforms prediction markets, with average squared prediction errors of 0.216 and 0.220, respectively. This is quite striking given its limited information set. It relies only on empirical distribution of test statistics of original findings, combined with the model outlined in Section 2.1. By contrast, prediction markets aggregate the judgments of many expert scientists, whose information sets may include detailed knowledge of the specific research question, broader field expertise, perceptions of author quality, and evidence from earlier replication studies. That a parsimonious statistical model can outperform this aggregation of expert judgments suggests that much of the signal relevant for replication is already embedded in the reported statistical evidence itself.

Overall, the validation results support using predictive power to evaluate replicability and suggest that it can provide a simple, low-cost alternative to prediction markets.

4. RESULTS

4.1 Replicability at Conventional Significance Thresholds

We turn now to the main question of interest: how likely are findings that just satisfy conventional significance thresholds to replicate when using the same sample size as the original study? Figure 2 plots the predictive power curve for experimental economics, psychology, and social science. The x -axis plots the observed p -value and

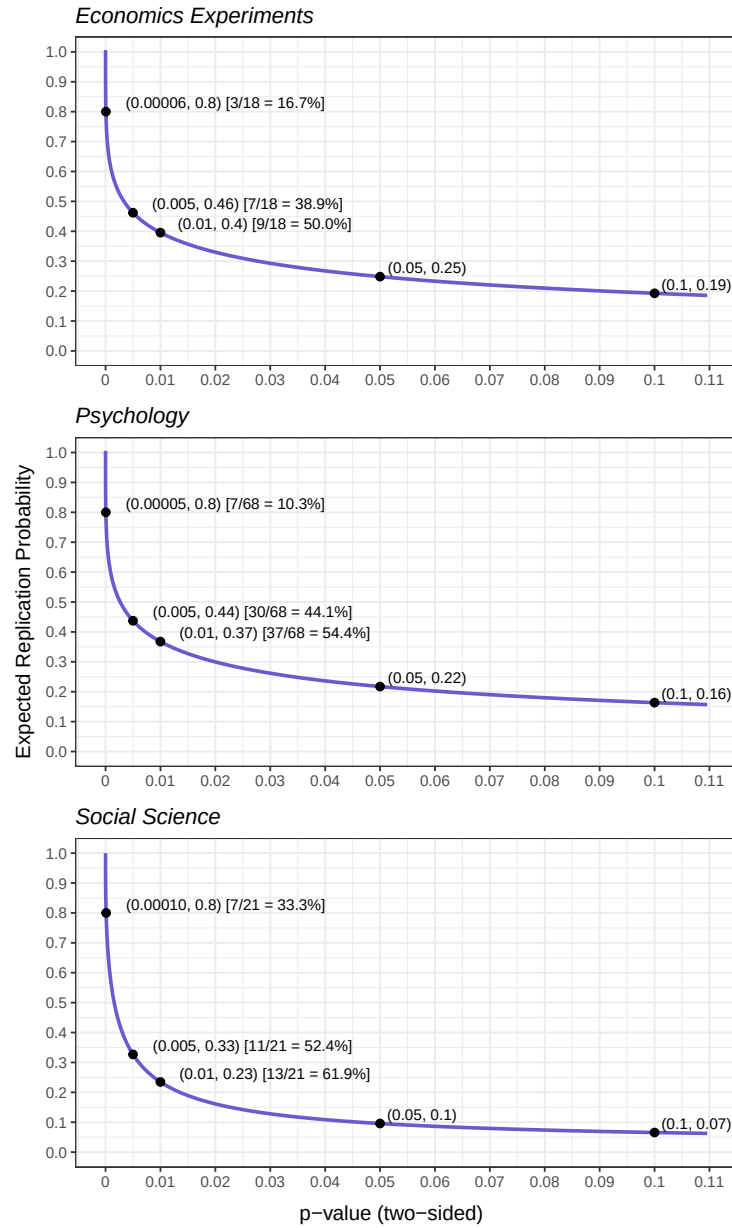
the y -axis plots $r(p)$, the replication probability at the 5% level. We find that replicability ranges between 0.1 and 0.25 across fields. Thus, a randomly chosen finding that just meets the most commonly used threshold for statistical significance has, at most, about a one-in-four chance of successful replication. At the more stringent 1% level, replication probabilities remain relatively low, ranging between 0.23 and 0.40 across fields. Thus, even at the most stringent threshold typically used in empirical research, the chance of replication remains less than one-half across fields.

It is important to note that these results come from a model that abstracts from many of the commonly proposed causes of low replicability, including researcher manipulation which misreports results, measurement error, and heterogeneity between original and replication studies (Ioannidis, 2005; Brodeur et al., 2016; Loken and Gelman, 2017; Elliott et al., 2022). Moreover, as discussed in Subsection 2.3, predictive power is invariant to selective reporting based on a finding’s test statistic. That the model nevertheless accurately predicts observed replication outcomes (Figure 1 and Table 1) suggests that these omitted factors are not necessary to explain the low replication rates observed in practice, at least in the literatures we examine.

Instead, the mechanism driving these results is the winner’s curse (Andrews et al., 2024). Specifically, just-significant estimates tend to reflect ‘favorable’ sampling variation, causing them to overstate true effects and making them less likely to replicate.¹² To quantify this overstatement, we estimate $\Pr(|\hat{z}| > |z| \mid p) = \Pr(|\hat{\beta}| > |\beta| \mid p)$, using the estimated distribution $\tilde{\pi}(z)$. At $p = 0.05$, we estimate that the observed effect overstates the magnitude of the true effect with probability 0.84 in experimental economics, 0.86 in psychology, and 0.94 in social science. Interestingly, at $p = 0.01$, these probabilities remain high at 0.83, 0.85, and 0.89, respectively. This occurs because moving from $p = 0.05$ to $p = 0.01$ both raises the observed estimate and shifts the posterior distribution of the true effect upward, leaving the probability that the former exceeds the latter relatively stable. Thus, while smaller p -values can raise replication probabilities somewhat, they do much less to reduce the chance that the point estimate overstates the magnitude of the true effect. The same mechanism also implies that many findings with $p = 0.05$, and even $p = 0.01$, are likely to perform

¹²Figures C2 and C3 illustrate this mechanism from two complementary perspectives. Conditional on $p = 0.05$, Figure C2 shows that predictive power is concentrated at low levels, and Figure C3 shows that the posterior distribution of the true effect places substantial mass well below the observed estimate $\hat{z} = 1.96$. Together, these results show that many just-significant findings combine relatively small true effects with favorable sampling variation.

FIGURE 2. Predictive Power Curves, by Replication Project



Notes: The figure plots estimated predictive power curves for experimental economics, psychology, and experimental social science, using the metastudy approach to estimate $\hat{\pi}(z)$. Predictive power (y -axis) is the probability that a replication with the same sample size as the original study produces an estimate that is statistically significant at the two-sided 5% level and has the same sign as the original estimate, conditional on the original study's two-sided p -value (x -axis). Square brackets report the percentage of original studies with significant p -values at or below the indicated threshold. Underlying data are from the replication projects of [Camerer et al. \(2016\)](#), [Open Science Collaboration \(2015\)](#), and [Camerer et al. \(2018\)](#), respectively.

poorly under replication criteria based on similarity in effect magnitude, such as the relative effect size $\hat{z}_r/\hat{z}$.

4.2 Interpreting Replication Outcomes

How should replication outcomes be interpreted in light of these results? We begin by clarifying what the results do not imply. Low replicability at conventional significance thresholds does not mean that most just-significant findings are ‘false’. Nor does an unsuccessful replication refute the original finding or necessarily indicate errors in the original study. Instead, replication outcomes should be interpreted as additional evidence that contributes to the cumulative evidence on an empirical claim.

To formalize this idea, consider a researcher interested in whether the true effect is ‘close to zero’. This framing meets practitioners “where they are,” given the ubiquity of null-hypothesis significance testing in applied research.¹³ Define a *near-null effect* as a true effect whose magnitude is small enough that a same-sized replication has less than a 5% probability of achieving 5% two-sided significance in the direction of the true effect. This corresponds to $|z| < 0.315$.¹⁴

Let $q_0(p) \equiv \Pr(H_0 \mid p)$ denote the probability of a near null conditional on the original p -value. Recall that $r(p) \equiv \Pr(R = 1 \mid p)$ is the probability of a successful replication, and define $r_0(p) \equiv \Pr(R = 1 \mid p, H_0)$ as the corresponding probability under a near null. By construction, $r_0(p) < 0.05$. Then it follows that

$$\begin{aligned} \Pr(H_0 \mid p, D = 1, R = 1) &= q_0(p) \left(\frac{r_0(p)}{r(p)} \right) \\ \Pr(H_0 \mid p, D = 1, R = 0) &= q_0(p) \left(\frac{1 - r_0(p)}{1 - r(p)} \right) \end{aligned} \tag{3}$$

The first term expresses the probability of a near null after a successful replication and the second after an unsuccessful replication. In both expressions, the first term is the probability of a near null based on the original evidence, and the second is the updating factor implied by the replication outcome. Figure 3 illustrates these

¹³For proposals to move beyond dichotomous significance testing, see [McShane et al. \(2019\)](#) and [Amrhein et al. \(2019\)](#).

¹⁴We focus on near-null effects because the estimated distribution of true effects is continuous and therefore assigns zero probability to an effect of exactly zero. Because the definition is stated in terms of the normalized effect z , rather than the underlying treatment effect β , the cutoff does not correspond to a common substantive effect size across studies and should be viewed as an operational definition.

expressions, plotting the probability of a near null effect before replication (dotted line), after a successful replication (green line), and after an unsuccessful replication (orange line).

A strikingly feature of Figure 3 is that, for most p -values, a successful replication sharply lowers the probability of a near null, whereas an unsuccessful replication raises it only modestly. For example, in experimental economics, a finding with $p = 0.05$ has a 0.16 probability of being a near null before replication. A successful replication reduces this probability sharply to 0.02, whereas an unsuccessful replication increases it only modestly to 0.21. This asymmetry arises because near nulls have, by construction, low replication probabilities, ranging from 0.025 to 0.05. A successful replication is therefore highly unlikely for near nulls and thus produces a large downward update in (3), whereas an unsuccessful replication remains relatively likely even when the effect is not close to zero, and therefore produces only a modest upward update. This asymmetry also appears in psychology and experimental social science, although near nulls are more likely in these fields for the same original p -value and replication outcome.

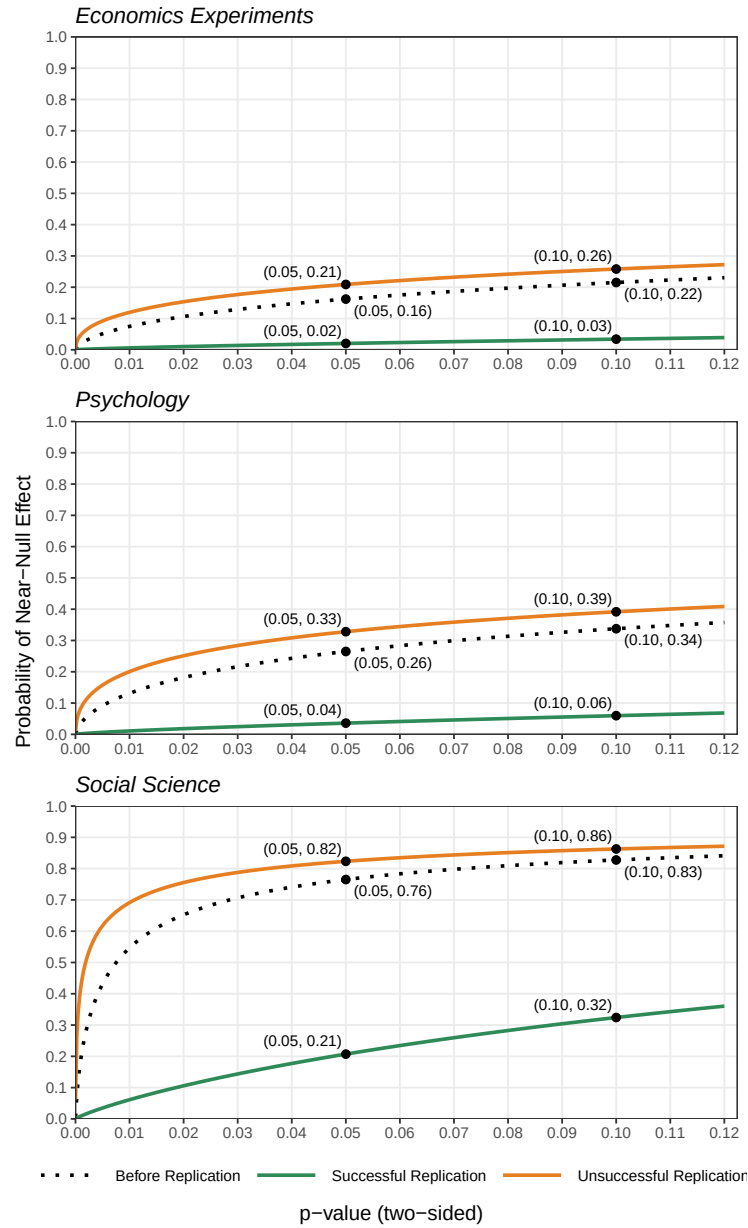
This asymmetry has implications for interpreting replication outcomes in practice: a successful replication can substantially strengthen evidence against a near null, whereas an unsuccessful replication generally provides only limited evidence in its favor. Thus, successful replications provide meaningful evidence for a nonzero effect, while unsuccessful replications should be generally not be interpreted as refutations of original result.

Taken together with the results of the previous subsection, these findings imply a more nuanced interpretation of low replication probabilities. While many significant findings have low replication probabilities and overstate the magnitude of the underlying effect, they are nevertheless unlikely to reflect near-null effects.

4.3 Additional Results

We highlight two additional implications of the predictive power curves. The first concerns how strong the statistical evidence in the original study must be to achieve a high probability of replication. For any replication probability target T , inverting the predictive power curve yields the p -value cutoff p^* satisfying $r(p^*) = T$. Lemma A.2 proves that, under mild regularity conditions, this cutoff exists and is unique. Figure 2 shows that achieving predictive power of at least 80% requires p -value cutoffs

FIGURE 3. Probability of Near Null Before and After Replication



Notes: The figure plots the probability of a near null before replication, after a successful replication, and after an unsuccessful replication, as a function of the original two-sided p -value. A near null is defined by $\{|z| < 0.315\}$, corresponding to a true effect whose magnitude implies less than a 5% probability that a replication achieves 5% two-sided significance in the direction of the true effect. A successful replication is statistically significant at the 5% level and has the same sign as the original estimate. Replications are assumed to use the same sample size as the original study. Underlying data are from [Camerer et al. \(2016\)](#), [Open Science Collaboration \(2015\)](#), and [Camerer et al. \(2018\)](#), respectively.

of 0.00006, 0.00005, and 0.0001 in economics, psychology, and the social sciences, respectively. Between 10% and 33% of findings in the respective samples meet these literature-specific benchmarks. Notably, these cutoffs are several orders of magnitude more stringent than conventional significance thresholds, and far below even the 0.005 threshold proposed in [Benjamin et al. \(2018\)](#). Nevertheless, we do not interpret these cutoffs as a recommendation to mechanically adopt more stringent thresholds. Although stricter thresholds would increase the average replicability of findings that meet them, they might also discard informative evidence, disadvantage settings where very large samples are infeasible, and intensify specification searching.

The second additional result concerns the convexity of the predictive power curves. In current empirical practice, moving from a p -value of 0.10 to 0.05 is commonly treated as a material gain in credibility, marking a transition from “marginal significance” to genuine “statistical significance.” However, [Figure 2](#) suggests that this transition does little to alter the underlying replicability of a finding: expected replication probabilities rise by only 3–6 percentage points, and remain at low levels. This implies that moving between conventional significance thresholds need not translate into meaningful improvements in replicability. A related implication concerns manipulation that moves reported p -values just across conventional significance thresholds. Because $r(p)$ changes little around $p = 0.05$, [Figure 2](#) suggests that such threshold-crossing manipulation alone is unlikely to generate large changes in expected replicability.

5. NONPARAMETRIC PREDICTIVE POWER CURVE

Thus far, the empirical analysis has imposed parametric assumptions about the distribution of true effects. In this section, we develop a nonparametric estimator of $r(p)$ that imposes no functional-form restrictions on $\pi(z)$. The nonparametric estimator requires large meta-samples that include many reported p -values and cannot be applied to the relatively small replication datasets analyzed above. Given these data requirements, we apply the estimator to the large dataset of experimental and quasi-experimental economics findings compiled in [Brodeur et al. \(2020\)](#).

Nonparametric identification of $r(p)$ is established formally in [Appendix B](#). Discontinuities in the density of published t -ratios identify changes in publication probability across significance thresholds. After adjusting for these changes, recovering

the distribution of true effects from the distribution of estimated t -ratios is a standard deconvolution problem with normally distributed errors.¹⁵ Below, we focus on the principal challenge of constructing a consistent estimator of $r(p)$. Proofs are in Appendix B.

5.1 Nonparametric Estimation

Suppose we observe a sample of absolute values of n published t -ratios, $\{|\hat{z}_i|\}_{i=1}^n$. Our aim is to estimate the symmetrized density $\tilde{\pi}(z) \equiv \frac{\pi(z) + \pi(-z)}{2}$, which determines the target parameter $r(p)$ and is identified from the distribution of $|\hat{z}|$. Following Example 1 of Carrasco and Florens (2011) and Faridani (2026), we can express $\tilde{\pi}(z)$ as a linear combination of known polynomial basis functions, χ_j , with unknown coefficients b_j :

$$\tilde{\pi}(z) = \sum_{j=0}^{\infty} b_j \chi_j(cv(p) - z), \quad (4)$$

where $\chi_j(z) \equiv \text{He}_j(z/\sqrt{2})/\sqrt{j!}$ is the normalized j th probabilists' Hermite polynomial.

Remark 1. *Estimating the full density $\tilde{\pi}(z)$ is severely ill-posed because many distributions of true effects can generate nearly indistinguishable distributions of observed t -ratios (Carroll and Hall, 1988; Fan, 1991). Distinguishing among them relies on higher-order coefficients b_j , which capture fine features of the data and are highly sensitive to sampling error. Despite this difficulty, the key insight underlying our estimator is that higher-order coefficients have diminishing influence on $r(p)$ because this estimand averages over $\tilde{\pi}(z)$ and is therefore relatively insensitive to its finer features. Consequently, imprecise estimates of $\tilde{\pi}(z)$ can yield accurate estimates of $r(p)$. A similar principle appears in Faridani (2026), but targeting $r(p)$ yields a new estimator with a distinct convergence rate.*

In practice, we truncate the infinite expansion and estimate only the first J_n coefficients. The cutoff J_n grows logarithmically with the sample size, allowing the estimator to capture finer features of $\tilde{\pi}(z)$ as more data become available. Formally, this amounts to regularizing the deconvolution problem using a spectral cutoff based

¹⁵See e.g. Carroll and Hall (1988); Fan (1991); Carrasco and Florens (2011); Faridani (2026).

on the singular value decomposition from Carrasco and Florens (2011).¹⁶ Substituting this truncated expansion into Equation (1) and simplifying yields the following estimator:

$$\hat{r}^{np}(p) \equiv 1 - \frac{\sum_{j=0}^{J_n} a_j \hat{b}_j}{\sum_{j=0}^{J_n} d_j \hat{b}_j} \quad (5)$$

where $\hat{b}_j$ estimates the coefficient b_j in (4) times a common constant (that cancels in the ratio). The known constants a_j and d_j capture the contribution of the j th basis function to the numerator and denominator, respectively, and can be computed numerically.¹⁷

To estimate the coefficients b_j consistently, we must account for selective publication of the observed t -ratios. As before, let $D = 1$ indicate that a finding is published, and suppose its publication probability depends on the observed t -ratio through the even selection function s_{θ_0} :

$$\Pr(D = 1 \mid \hat{z}) = s_{\theta_0}(\hat{z}) \quad (6)$$

where the functional form of s_{θ} is known but the parameter vector $\theta_0 \in \Theta$ is unknown.

The properties of the Hermite basis allow each coefficient b_j to be expressed as a moment of the unselected distribution of estimated t -ratios. Given a consistent estimator $\hat{\theta}_n$ of the publication-bias parameters, we follow Faridani (2026) and estimate this moment by weighting each published t -ratio by the inverse of its estimated publication probability:

$$\hat{b}_j \equiv \frac{2^{j/2}}{n} \sum_{i=1}^n \frac{\psi_j(cv(p) - |\hat{z}_i|) \varphi(cv(p) - |\hat{z}_i|) + \psi_j(cv(p) + |\hat{z}_i|) \varphi(cv(p) + |\hat{z}_i|)}{s_{\hat{\theta}_n}(\hat{z}_i)} \quad (7)$$

where $\varphi(\cdot)$ is the standard normal density, $\psi_j(z) \equiv \text{He}_j(z) / \sqrt{j!}$ and $cv(p) \equiv \Phi^{-1}(1 - p/2)$ is the standard normal critical value corresponding to the two-sided p -value.

Next we formalize our assumptions about the sample. In particular, test statistics, true effects, and true standard errors are allowed to be arbitrarily dependent within

¹⁶See Example 1 of Carrasco and Florens (2011) for the singular value decomposition. The Gibbs fluctuations incurred by spectral cutoff do not prevent consistency because they are integrated away when we calculate $r(p)$ from $\tilde{\pi}(z)$.

¹⁷More specifically, $a_j \equiv \int_{\mathbb{R}} \varphi(u) e^{-u^2/2} \Phi(c - cv(p) + \sqrt{2}u) \psi_j(u) du$, and $d_j \equiv \int_{\mathbb{R}} \varphi(u) e^{-u^2/2} \psi_j(u) du$.

an article, but must be independent across articles.

Assumption 1. *The following all hold:*

1. *Let $|\hat{z}_1|, \dots, |\hat{z}_n|$ be an identically distributed sample from the distribution of $|\hat{z}|$ conditional on publication, i.e. $D = 1$.*
2. *Sets of $\hat{z}_i$ drawn from different articles are independent.*
3. *There is a universal constant $L \geq 1$ such that each article contributes no more than L observations to the sample.*

Assumption 1 permits rich dependence among findings reported in the same article. For example, estimates from different outcomes or specifications may have correlated sampling errors and may correspond to correlated underlying true effects.¹⁸

Theorem 1 shows that, if the publication-bias parameters θ_0 are known or consistently estimated by $\hat{\theta}_n$, then the estimator $\hat{r}^{np}(p)$ defined in (5), with coefficients given by (7), is consistent for $r(p)$.¹⁹

Theorem 1. *Let the sample $|\hat{z}_1|, \dots, |\hat{z}_n|$ satisfy Assumption 1. Suppose that the density $\pi(z)$ exists and has bounded height. Assume that s_θ is even and $\inf_{\theta \in \Theta, t \in \mathbb{R}} s_\theta(t) > 0$. Let $J_n = \left\lceil \frac{2}{3 \log 2} \log(An) \right\rceil$ for any $A > 0$. Assume further that an estimator $\hat{\theta}_n$ satisfies:*

$$\sup_{t \in \mathbb{R}} n^{1/3} \left| \frac{1}{s_{\hat{\theta}_n}(t)} - \frac{1}{s_{\theta_0}(t)} \right| = \mathcal{O}_p(1)$$

Then, for every $p \in (0, 1)$,

$$\hat{r}^{np}(p) \xrightarrow{p} r(p)$$

We complete the estimation procedure by specifying and estimating the publication-bias function. To allow for flexible forms of publication bias, we model the publication

¹⁸Separately, the Andrews and Kasy (2019) model restricts selection: a finding's reporting probability may depend arbitrarily on its own test statistic, but not generally on the full vector of test statistics in the article. For example, reporting each outcome based on its own significance is allowed, whereas reporting only the most significant outcome is not.

¹⁹This result is a special case of the more general Theorem B.2 in Appendix B, which allows the replication and original sample sizes to differ.

probability as a step function with M known critical values $\{v_m\}_{m=1}^M$ where the publication probability of the most-significant $\hat{z}$ is normalized to one:

$$s_\theta(\hat{z}) = \theta_{(1)} \mathbf{1}\{|\hat{z}| \leq v_1\} + \sum_{m=2}^M \theta_{(m)} \mathbf{1}\{|\hat{z}| \in (v_{m-1}, v_m]\} + \mathbf{1}\{|\hat{z}| > v_M\} \quad (8)$$

Given the known step locations, we estimate the publication-bias parameters using the caliper ratio estimator of [Faridani \(2026\)](#). The estimator compares the numbers of published t -ratios within narrow intervals on either side of each critical value. Discontinuities in these counts identify changes in publication probability, from which the step heights can be recovered. The resulting estimates are uniformly consistent for s_θ and can be used in Theorem 1. For further details, see [Faridani \(2026\)](#).

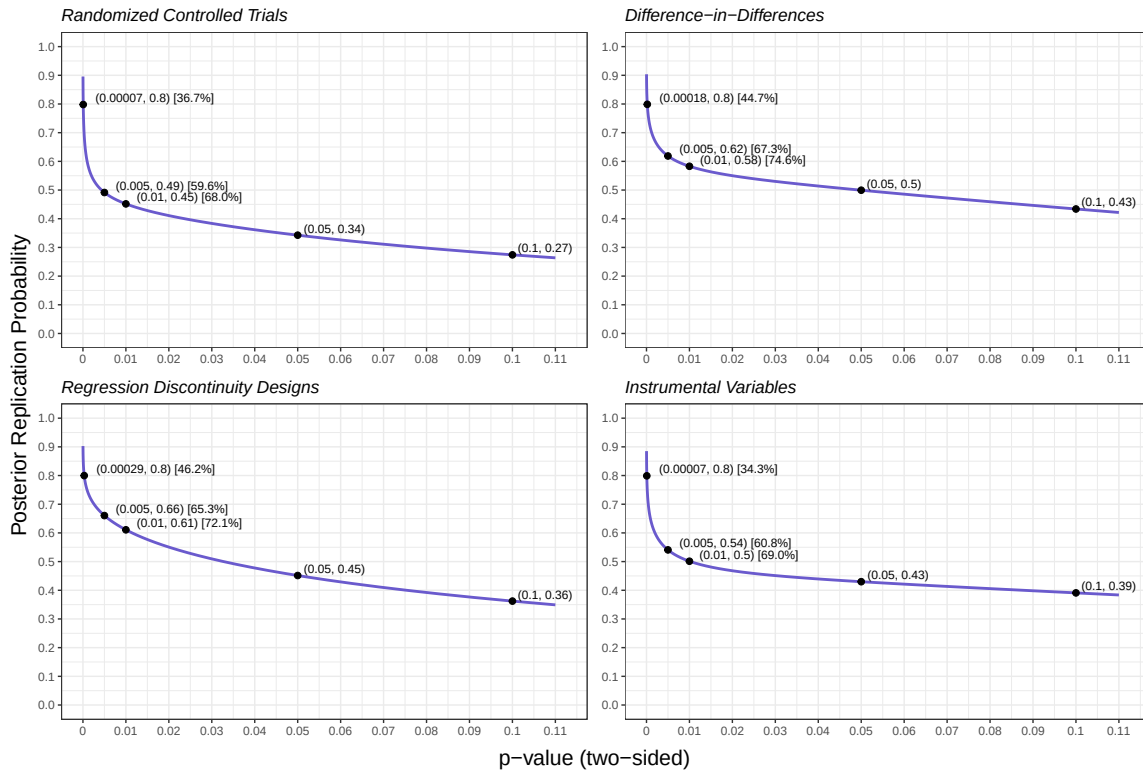
5.2 Application

We apply the nonparametric estimator to the dataset in [Brodeur et al. \(2020\)](#), which consists of 21,740 t -ratios testing main hypotheses in 684 articles published in top economics journals. Each t -ratio corresponds to the test of a causal inference hypothesis where the study methodology is either a randomized controlled trial (RCT), Regression Discontinuity Design (RDD), Difference-in-Differences (DID), or Instrumental Variables (IV).²⁰ Papers can contain multiple estimates. Following [Kranz and Pütz \(2022\)](#), we de-round the t -ratios before estimation because rounding reported coefficients and standard errors can generate artificial bunching around conventional significance thresholds. The publication bias function, $s_\theta(\cdot)$, is specified to have steps at the three conventional two-sided critical values 1.645, 1.96, and 2.576. Estimates for the publication probability function are presented in Appendix E.

To make the replication exercise concrete for observational studies, consider an RDD using the American Community Survey (ACS) to estimate the effect of Medicare eligibility at age 65 on health-insurance coverage. Suppose the original study uses one annual ACS Public Use Microdata Sample, which contains approximately 1% of the U.S. population. Replication can then be viewed as drawing an independent representative sample of the same size from the same population and policy environment and applying the identical RDD specification.

²⁰For IV estimates, the normal approximation may be less accurate when instruments are weak; we therefore interpret the IV results with this additional qualification.

FIGURE 4. Nonparametric Predictive Power Curves, by Research Design in Economics



Notes: The figure plots nonparametric estimates of the predictive power curve by research design in economics. Predictive power (y -axis) is the probability that a replication with the same sample size as the original study produces an estimate that is statistically significant at the two-sided 5% level and has the same sign as the original estimate, conditional on the original study's two-sided p -value (x -axis). Square brackets report the percentage of original studies with significant p -values at or below the indicated threshold. Underlying data are from [Brodeur et al. \(2020\)](#).

Figure 4 presents the nonparametric predictive power curves for the four research designs. An RCT finding with a p -value of 0.05 has an estimated replication probability of 0.34, modestly exceeding the parametric estimate of 0.25 for experimental economics. Predictive power at $p = 0.05$ is notably higher for the observational designs, ranging between 0.43 and 0.50, although, in absolute terms, replication probabilities remain relatively low. Similar to the parametric analysis in Figure 2, extremely small p -values are required to achieve predictive power of at least 80%.

A notable difference is that, among statistically significant findings, highly replicable findings are substantially more common in the nonparametric sample. The share of significant findings with predictive power of at least 0.8 is 36.7% for RCTs, 44.7% for DID, 46.2% for RDD, and 34.3% for IV, compared with 16.7% in the

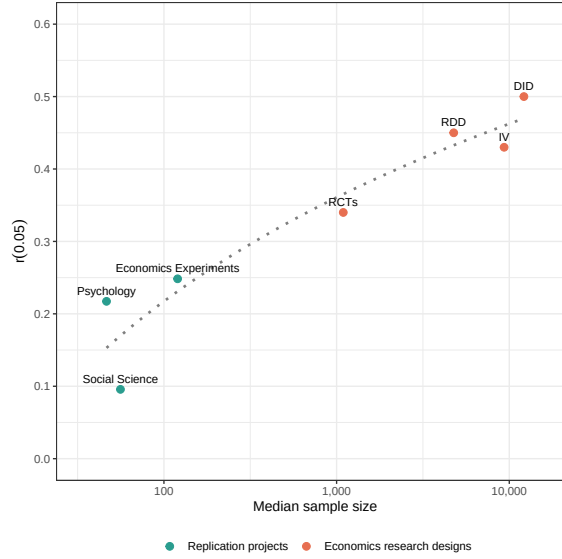
experimental-economics replication sample. Thus, although just-significant findings remain difficult to replicate, stronger statistical evidence is considerably more common in these economics literatures.

Differences in replicability across the seven applications are closely related to differences in sample size. Figure 5 plots median sample size against predictive power at $p = 0.05$. It shows that predictive power is approximately linear in the logarithm of median sample size, with the applications lying close to the fitted line. Higher predictive power in observational studies is consistent with their much larger samples. Similarly, economics RCTs—which include relatively large field experiments—have higher replicability than smaller laboratory studies in experimental economics. These patterns are consistent with replicability being determined primarily by the distribution of normalized true effects, $\tilde{\pi}(z)$, as emphasized in Section 4. For a given underlying effect, a larger sample reduces the standard error and increases the normalized true effect, shifting $\tilde{\pi}(z)$ away from zero and increasing predictive power.

These results offer an interesting perspective on the relationship between research practices and replicability. A literature with more evidence of selective reporting or p -hacking may nevertheless be more replicable. For example, Brodeur et al. (2020) use discontinuities at conventional significance thresholds as evidence that IV studies—and, to a lesser extent, DID studies—exhibit substantially more selective reporting or p -hacking than RCTs. Yet Figure 5 shows that just-significant RCT findings have much lower expected replicability than IV and DID findings, consistent with their smaller sample sizes. We note that this comparison concerns replicability rather than internal validity. For example, quasi-experimental results might still be compromised by violations of exclusion restrictions, failed pretrends, or other design-specific identification problems.

Overall, the nonparametric analysis extends the earlier parametric results to a broader range of research designs while allowing for a flexible distribution of true effects. Expected replicability at conventional significance thresholds is higher for observational designs than for experiments, but remains low in absolute terms across all applications.²¹

²¹The additional analyses in Figure 3 are not feasible with the nonparametric estimator because they require estimation of $\tilde{\pi}(z)$ itself. While the estimator of $r(p)$ converges relatively quickly, the nonparametric estimator of $\tilde{\pi}(z)$ converges slowly and is not guaranteed to yield a valid density.

FIGURE 5. Median Sample Size and Expected Replicability at $p = 0.05$ 

Notes: The figure plots the expected replication probability at an original two-sided p -value of 0.05 against the median sample size for each application. Replication project data are from [Camerer et al. \(2016\)](#), [Open Science Collaboration \(2015\)](#), and [Camerer et al. \(2018\)](#). For RCTs, IV, RDD, and DID, data are from [Brodeur et al. \(2020\)](#), with median sample sizes calculated from the nonmissing values. The dotted line plots the OLS fitted values from $r_i(0.05) = \alpha + \beta \log_{10}(\text{Median } N_i) + \varepsilon_i$.

6. RECOMMENDATIONS

This section offers three recommendations for researchers and replicators. Recommendations are broadly applicable to replications evaluated using either binary significance criteria or effect sizes.

6.1 Cumulative Evidence

Recommendation 1. *Treat evidence as cumulative, not binary. Successful replications can provide strong evidence against the null, whereas insignificant replications do not, by themselves, refute the original finding or imply flaws in the original study.*

Replications based on statistical significance do not deliver a definitive conclusions about the original claim, confirming it if “successful” and refuting it otherwise. For interpreting replication outcomes, the analysis in Subsection 4.2 reveals an important asymmetry: a successful replication can substantially strengthen evidence against a near null result, whereas an unsuccessful replication generally provides only

limited evidence in its favor. Rather than viewing individual replications as providing decisive evidence, they are better understood as contributing to the cumulative body of evidence needed to distinguish persistent empirical relationships from isolated findings driven by sampling variation. The ultimatum-game literature provides an instructive example, where a large number of replications have established robust empirical regularities (Coffman and Niederle, 2015). Treating scientific evidence as cumulative aligns with arguments developed by Popper (1934) in *The Logic of Scientific Discovery*: “We do not take even our own observations quite seriously, or accept them as scientific observations, until we have repeated and tested them. Only by such repetitions can we convince ourselves that we are not dealing with a mere isolated ‘coincidence’, but with events which, on account of their regularity and reproducibility, are in principle inter-subjectively testable.”

6.2 Replication Design

Recommendation 2. *Use the largest feasible replication sample to better distinguish between the null and alternative hypotheses. Treat nominal power targets with caution when power calculations substitute the estimated effect for the true effect: expected replication rates will generally be below the nominal target, even without selective reporting or researcher manipulation.*

The same-sample-size benchmark used in this paper evaluates replicability under the original research design; it is not intended as a recommendation for replication design. Instead, we recommend that replicators use the largest feasible sample, which maximizes predictive power. This is because larger samples make replication outcomes more informative: significance provides stronger evidence against the null, whereas insignificance provides stronger evidence in its favor.²²

When evidence informs a concrete decision, the replication sample size should balance the value of more precise evidence against the cost of collecting additional data. In principle, this requires weighing sampling costs against the consequences of adopting an ineffective intervention or rejecting a beneficial one.

Finally, we recommend caution when using the conventional approach of choosing a replication sample size to detect the original effect estimate with, for example, 80%

²²Formally, consider the updating factors in equation (3). Increasing the replication sample size lowers $\left(\frac{r_0(p)}{r(p)}\right)$ following significance and raises $\left(\frac{1-r_0(p)}{1-r(p)}\right)$ following insignificance.

or 90% power. Such calculations treat the original estimate as exactly equal to the true effect and ignore its sampling uncertainty. Due to the nonlinearity of the power function, Vu (2024) shows that this approach delivers expected power strictly below the nominal target, even absent selective reporting or researcher manipulation.²³ Of course, outcomes from such replications can still be used to rationally update beliefs about the null. Our concern is rather that stated replication-rate targets may create the mistaken impression that they are attainable absent research distortions, which is not the case.

6.3 Encourage and Reward Replications

Recommendation 3. *Encourage the publication and citation of replications to establish empirical regularities. Preregistration and pre-analysis plans remain valuable for limiting selective reporting and other forms of researcher discretion, but significant findings can still have low replicability even in their absence.*

Our finding of low replicability at conventional thresholds reinforces recent calls for greater incentives to conduct, publish, and recognize replications (Nosek et al., 2012; Coffman and Niederle, 2015; Brodeur et al., 2023). As shown in Subsection 4.2, successful replications can generate substantial information about the null hypothesis and hence provide considerable value for subsequent researchers and decision-makers.

Despite this, replications offer less novelty and professional reward than original studies. To address this imbalance, Coffman et al. (2017) make two proposals. First, leading journals could publish short reports summarizing the replication evidence for influential studies, including replications otherwise buried within broader papers. Second, researchers could cite replications alongside original studies, ensuring that credit and attention reflect the cumulative evidence rather than the first published result alone. Complementing these proposals, the Replication Games of Brodeur et al. (2023) use team-based workshops and joint meta-papers to coordinate replication efforts and reward participants.

Finally, it is important to emphasize that preregistration, pre-analysis plans, and registered reports can limit selective reporting and other forms of researcher discre-

²³The intuition is as follows. Let T denote the nominal power target and $g(\cdot)$ denote the probability of replication. Suppose $\hat{\beta} \sim N(\beta, \sigma^2)$. Because $g(\cdot)$ is locally concave, Jensen's inequality implies that $\mathbb{E}[g(\hat{\beta})] < g(\mathbb{E}[\hat{\beta}]) = g(\beta) = T$. Thus, even though the original estimates are unbiased, expected replication power is below the nominal target T .

tion, but low replicability can persist even in their absence. Hence, improving replicability and limiting selective reporting are distinct objectives. Three observations underscore this point. First, expected replicability at a given p -value is invariant to selective reporting based on a finding’s own test statistic and instead determined by the underlying distribution of true effects (Subsection 2.3). Second, Section 5 finds higher replicability in observational studies than RCTs, despite stronger evidence of selective reporting or p -hacking in the former (Brodeur et al., 2020; Kranz and Pütz, 2022). Third, the impact of imposing more stringent publication thresholds would be to mechanically increase replicability, since this favors findings with higher statistical power. In other words, increasing selectivity *improves* replicability. For these reasons, reforms to reduce selective reporting and promote systematic replication should be pursued as complementary objectives.

7. CONCLUSION

In the empirical sciences, statistically significant findings are often treated as evidence that an effect. This paper asks how likely such findings are to remain significant in an independent replication of the same sample size. Using an empirically validated measure of predictive power across fields and research designs, we find that replication probabilities at conventional thresholds are generally low. These results arise without selective reporting and instead reflect the winner’s curse: statistically significant estimates often combine small true effects with favorable sampling variation, causing them to overstate the magnitude of the underlying effect and making significance unstable across samples. Robust scientific conclusions therefore require aggregating evidence across independent studies to establish whether an effect persists and to obtain more reliable estimates of its magnitude.

REFERENCES

- Altmejd, Adam, Anna Dreber, Eskil Forsell et al.**, “Predicting the Replicability of Social Science Lab Experiments,” *PLoS ONE*, 2019, 14 (12).
- Amrhein, Valentin, David Trafimow, and Sander Greenland**, “Inferential Statistics as Descriptive Statistics: There Is No Replication Crisis if We Don’t Expect Replication,” *The American Statistician*, 2019, 73 (1), 262–270.

- Andrews, Isaiah and Maximilian Kasy**, “Identification of and Correction for Publication Bias,” *American Economic Review*, 2019, *109* (8), 2766–2794.
- , **Toru Kitagawa**, and **Adam McCloskey**, “Inference on Winners,” *The Quarterly Journal of Economics*, 02 2024, *139* (1), 305–358.
- Benjamin, Daniel J., James O. Berger, Magnus Johannesson et al.**, “Redefine Statistical Significance,” *Nature Human Behaviour*, 2018, *2*, 6–10.
- Brodeur, Abel, Anna Dreber, Fernando Hoces de la Guardia, and Edward Miguel**, “Replication Games: How to Make Reproducibility Research More Systematic,” *Nature*, 2023, *621* (7980), 684–686.
- , **Mathias Lé, Marc Sangnier, and Yanos Zylberberg**, “Star Wars: The Empirics Strike Back,” *American Economic Journal: Applied Economics*, 2016, *8* (1), 1–32.
- , **Nikolai Cook, and Anthony Heyes**, “Methods Matter: p-Hacking and Publication Bias in Causal Analysis in Economics,” *American Economic Review*, 2020, *110* (11), 3634–3660.
- Camerer, Colin F., Anna Dreber, Eskil Forsell et al.**, “Evaluating Replicability of Laboratory Experiments in Economics,” *Science*, 2016, *351* (6280), 1433–1436.
- , —, **Felix Holzmeister et al.**, “Evaluating the replicability of social science experiments in Nature and Science between 2010 and 2015,” *Nature Human Behaviour*, 2018, *2* (9), 637–644.
- Card, David and Alan B. Krueger**, “Time-Series Minimum-Wage Studies: A Meta-Analysis,” *American Economic Review*, 1995, *85* (2), 238–243.
- Carrasco, Marine and Jean-Pierre Florens**, “A SPECTRAL METHOD FOR DECONVOLVING A DENSITY,” *Econometric Theory*, 2011, *27* (3), 546–581.
- Carroll, Raymond J. and Peter Hall**, “Optimal Rates of Convergence for Deconvolving a Density,” *Journal of the American Statistical Association*, 1988, *83* (404), 1184–1186.
- Coffman, Lucas C. and Muriel Niederle**, “Pre-Analysis Plans Have Limited Upside, Especially Where Replications Are Feasible,” *Journal of Economic Perspectives*, 2015, *29* (3), 81–98.
- , —, and **Alistair J. Wilson**, “A Proposal to Organize and Promote Replications,” *American Economic Review*, 2017, *107* (5), 41–45.

- Cohen, Jacob**, *Statistical Power Analysis for the Behavioral Sciences*, 2 ed., Hillsdale, NJ: Lawrence Erlbaum Associates, 1988.
- DellaVigna, Stefano and Elizabeth Linos**, “RCTs to Scale: Comprehensive Evidence From Two Nudge Units,” *Econometrica*, 2022, *90* (1), 81–116.
- , **Devin Pope, and Eva Vivalt**, “Predict Science to Improve Science,” *Science*, 2019, *336* (6464), 428–429.
- , **Nicholas Otis, and Eva Vivalt**, “Forecasting the Results of Experiments: Piloting an Elicitation Strategy,” *AEA Papers and Proceedings*, 2020, *110*, 75–79.
- Dreber, Anna, Thomas Pfeiffer, Johan Almenberg et al.**, “Using Prediction Markets to Estimate the Reproducibility of Scientific Research,” *Proceedings of the National Academy of Sciences of the United States of America*, 2015, *112* (50), 15343–15347.
- Elliott, Graham, Nikolay Kudrin, and Kaspar Wüthrich**, “Detecting p-Hacking,” *Econometrica*, 2022, *90* (2), 887–906.
- Fan, Jianqing**, “GLOBAL BEHAVIOR OF DECONVOLUTION KERNEL ESTIMATES,” *Statistica Sinica*, 1991, *1* (2), 541–551.
- Faridani, Stefan**, “Testing for underpowered literatures,” *Journal of Econometrics*, 2026, *257*, 106312.
- Franco, Annie, Neil Malhotra, and Gabor Simonovits**, “Publication Bias in the Social Sciences: Unlocking the File Drawer,” *Science*, 2014, *345* (6203), 1502–1505.
- Gordon, Michael, Domenico Viganola, Anna Dreber, Magnus Johannesson, and Thomas Pfeiffer**, “Predicting replicability—Analysis of survey and prediction market data from large-scale forecasting projects,” *PLOS ONE*, 2021, *16* (4).
- Hoenig, John M. and Dennis M. Heisey**, “The Abuse of Power: The Pervasive Fallacy of Power Calculations for Data Analysis,” *The American Statistician*, 2001, *55* (1), 19–24.
- Indritz, Jack**, “An Inequality for Hermite Polynomials,” *Proceedings of the American Mathematical Society*, 1961, *12* (6), 981–983.
- Ioannidis, John P.A.**, “Why Most Published Research Findings Are False,” *PLoS Medicine*, 2005, *2* (8).

- , **T. D. Stanley**, and **Hristos Doucouliagos**, “The Power of Bias in Economics Research,” *The Economic Journal*, 2017, 127 (605), 236–265.
- Kranz, Sebastian** and **Peter Pütz**, “Methods Matter: p-Hacking and Publication Bias in Causal Analysis in Economics: Comment,” *American Economic Review*, 2022, 112 (9), 3124–3136.
- Lang, Kevin**, “How Credible Is the Credibility Revolution?,” *Journal of Labor Economics*, 2025, 43 (2), 635–663.
- Loken, Eric** and **Andrew Gelman**, “Measurement Error and the Replication Crisis,” *Science*, 2017, 355 (6325), 584–585.
- Maxwell, Scott E.**, **Michael Y. Lau**, and **George S. Howard**, “Is Psychology Suffering from a Replication Crisis? What Does “Failure to Replicate” Really Mean? ,” *American Psychologist*, 2015, 70 (6), 487–498.
- McShane, Blakeley B.**, **David Gal**, **Andrew Gelman et al.**, “Abandon Statistical Significance,” *The American Statistician*, 2019, 73 (1), 235–245.
- Nosek, Brian A.**, **Jeffrey R. Spies**, and **Matt Motyl**, “Scientific Utopia: II. Restructuring Incentives and Practices to Promote Truth over Publishability,” *Perspectives on Psychological Science*, 2012, 7 (6), 615–631.
- Open Science Collaboration**, “Estimating the Reproducibility of Psychological Science,” *Science*, 2015, 349 (6251).
- Popper, Karl R.**, *The Logic of Scientific Discovery*, London: Routledge, 1934. English translation first published in 1959; this edition published in 2002. Originally published as *Logik der Forschung* (Vienna: Julius Springer, 1934).
- van Zwet, Erik W.** and **Steven N. Goodman**, “How large should the next study be? Predictive power and sample size requirements for replication studies,” *Statistics in Medicine*, 2022, 41 (16), 3090–3101.
- Vu, Patrick**, “Why are replication rates so low?,” *Journal of Econometrics*, 2024, 245 (1).
- Wald, Abraham**, *Statistical Decision Functions*, New York: John Wiley & Sons, 1950.

A. PROOFS

Proof of Lemma 1: Let $f_{z||\hat{z}|,D}(\cdot | |\hat{z}|, 1)$ denote the conditional density of z given $|\hat{z}|$ and publication, $D = 1$. Then

$$\Pr(R = 1 | |\hat{z}| = x, D = 1) = \int_{\mathbb{R}} \Pr(R = 1 | |\hat{z}| = x, z, D = 1) f_{z||\hat{z}|,D}(z | x, 1) dz$$

Conditional on $|\hat{z}| = x$ and z , the original test statistic can take either value $\hat{z} = x$ or $\hat{z} = -x$. Replication success requires statistical significance in the same direction as the original estimate. Using the Law of Total Probability,

$$\begin{aligned} \Pr(R = 1 | |\hat{z}| = x, z, D = 1) &= \Pr(\hat{z}_r > 1.96 | z) \Pr(\hat{z} = x | |\hat{z}| = x, z, D = 1) \\ &\quad + \Pr(\hat{z}_r < -1.96 | z) \Pr(\hat{z} = -x | |\hat{z}| = x, z, D = 1) \\ &= \frac{[1 - \Phi(1.96 - z)]s(x)\varphi(x - z) + \Phi(-1.96 - z)s(-x)\varphi(x + z)}{s(x)\varphi(x - z) + s(-x)\varphi(x + z)} \\ &= \frac{[1 - \Phi(1.96 - z)]\varphi(x - z) + \Phi(-1.96 - z)\varphi(x + z)}{\varphi(x - z) + \varphi(x + z)} \end{aligned}$$

Conditional on $|\hat{z}| = x$, the original test statistic is either x or $-x$. The second conditional probability in each product gives the probability of each realization conditional on z and publication. The final equality follows from the symmetry of the publication function: $s(x) = s(-x)$.

Next, Bayes' rule implies

$$\begin{aligned} f_{z||\hat{z}|,D}(z | x, 1) &= \frac{f_{D||\hat{z}|,z}(1 | x, z) f_{|\hat{z}||z}(x | z) \pi(z)}{\int_{\mathbb{R}} f_{D||\hat{z}|,z'}(1 | x, z') f_{|\hat{z}||z'}(x | z') \pi(z') dz'} \\ &= \frac{s(x) [\varphi(x - z) + \varphi(x + z)] \pi(z)}{\int_{\mathbb{R}} s(x) [\varphi(x - z') + \varphi(x + z')] \pi(z') dz'} \\ &= \frac{[\varphi(x - z) + \varphi(x + z)] \pi(z)}{\int_{\mathbb{R}} [\varphi(x - z') + \varphi(x + z')] \pi(z') dz'} \end{aligned}$$

where $f_{D||\hat{z}|,z}(1 | x, z) = s(x)$ since publication depends on $\hat{z}$ through $s(\hat{z})$ and symmetry implies $s(x) = s(-x)$.

Combining the two above expressions and using $cv(p) \equiv |\hat{z}| = \Phi^{-1}(1 - p/2)$ gives

$$\Pr(R = 1 | p, D = 1) = \frac{\int_{\mathbb{R}} [1 - \Phi(1.96 - z)]\varphi(cv(p) - z)\pi(z) dz + \int_{\mathbb{R}} \Phi(-1.96 - z)\varphi(cv(p) + z)\pi(z) dz}{\int_{\mathbb{R}} [\varphi(cv(p) - z) + \varphi(cv(p) + z)] \pi(z) dz}$$

For the second term in the numerator, let $u = -z$. Then

$$\int_{\mathbb{R}} \Phi(-1.96 - z) \varphi(cv(p) + z) \pi(z) dz = \int_{\mathbb{R}} [1 - \Phi(1.96 - u)] \varphi(cv(p) - u) \pi(-u) du$$

Similarly,

$$\int_{\mathbb{R}} \varphi(cv(p) + z) \pi(z) dz = \int_{\mathbb{R}} \varphi(cv(p) - u) \pi(-u) du$$

Recall the symmetrized distribution is defined as $\tilde{\pi}(z) \equiv \frac{\pi(z) + \pi(-z)}{2}$. Relabelling u as z , we obtain

$$\begin{aligned} \Pr(R = 1 \mid p, D = 1) &= \frac{\int_{\mathbb{R}} [1 - \Phi(1.96 - z)] \varphi(cv(p) - z) [\pi(z) + \pi(-z)] dz}{\int_{\mathbb{R}} \varphi(cv(p) - z) [\pi(z) + \pi(-z)] dz} \\ &= \frac{\int_{\mathbb{R}} [1 - \Phi(1.96 - z)] \varphi(cv(p) - z) \tilde{\pi}(z) dz}{\int_{\mathbb{R}} \varphi(cv(p) - z) \tilde{\pi}(z) dz} \end{aligned}$$

which completes the proof. $\square$

Lemma A.1 (Predictive Power Under the Fractional Power Rule). *Let $s(\cdot)$ be symmetric about zero and strictly positive. Suppose the replication standard error is chosen according to the fractional power rule with fraction $\psi \in (0, 1]$ of the original effect and nominal target power $1 - \eta$:*

$$\sigma_r = \frac{\psi |\hat{z}|}{1.96 - \Phi^{-1}(\eta)} \sigma$$

Then predictive power is

$$r(p) = \frac{\int_{\mathbb{R}} \left[1 - \Phi \left(1.96 - z \frac{1.96 - \Phi^{-1}(\eta)}{\psi cv(p)} \right) \right] \varphi(cv(p) - z) \tilde{\pi}(z) dz}{\int_{\mathbb{R}} \varphi(cv(p) - z) \tilde{\pi}(z) dz} \quad (9)$$

where $cv(p) \equiv \Phi^{-1}(1 - p/2)$ and $\tilde{\pi}(z) \equiv \frac{\pi(z) + \pi(-z)}{2}$. The common power rule is the special case $\psi = 1$.

Proof. Let $x \equiv |\hat{z}|$. Under the fractional power rule,

$$z_r \equiv \frac{\beta}{\sigma_r} = \frac{\beta}{\sigma} \frac{\sigma}{\sigma_r} = z \frac{1.96 - \Phi^{-1}(\eta)}{\psi x}$$

Hence, conditional on z and $|\hat{z}| = x$, the replication test statistic, $\hat{z}_r$ has mean $z \frac{1.96 - \Phi^{-1}(\eta)}{\psi x}$.

Using identical arguments to the proof of Lemma 1, we obtain Equation (9). Setting $\psi = 1$ gives the common power rule as a special case. $\square$

Lemma A.2 (Existence and Uniqueness of p -value Cutoff for Target Power). *Let $s(\cdot)$ be symmetric about zero and strictly positive, and assume that $\tilde{\pi}(\cdot)$ is non-degenerate. Define*

$\underline{r} \equiv \lim_{p \rightarrow 1} r(p)$ and $\bar{r} \equiv \lim_{p \rightarrow 0} r(p)$. Then, for every target $T \in (\underline{r}, \bar{r})$, there exists a unique $p^ \in (0, 1)$ such that $r(p^*) = T$.*

Furthermore, if $\tilde{\pi}(\cdot)$ has unbounded support above, then $\bar{r} = 1$. Consequently, every target replication probability $T \in (\underline{r}, 1)$ is attained by a unique p -value cutoff. Finally, $\underline{r} > 1 - \Phi(1.96) = 0.025$.

Proof. Let $x \equiv cv(p) = \Phi^{-1}(1 - p/2)$ and define

$$g(x) \equiv \frac{\overbrace{\int_{\mathbb{R}} [1 - \Phi(1.96 - z)] \varphi(x - z) \tilde{\pi}(z) dz}^{\equiv N(x)}}{\underbrace{\int_{\mathbb{R}} \varphi(x - z) \tilde{\pi}(z) dz}_{\equiv D(x)}}$$

By Lemma 1, $r(p) = g(cv(p))$.

Continuity of $g(x)$ follows from the dominated convergence theorem. In particular, the integrands defining $N(x)$ and $D(x)$ are continuous in x and bounded in absolute value by a constant multiple of $\tilde{\pi}(z)$. Hence, $N(x)$ and $D(x)$ are continuous, and since $D(x) > 0$, so is $g(x)$.

Next, define

$$q_x(z) \equiv \frac{\varphi(x - z) \tilde{\pi}(z)}{D(x)}$$

For any $x_2 > x_1$, the likelihood ratio is

$$\frac{q_{x_2}(z)}{q_{x_1}(z)} = \frac{D(x_1) \varphi(x_2 - z)}{D(x_2) \varphi(x_1 - z)} = \frac{D(x_1)}{D(x_2)} \exp \left\{ (x_2 - x_1)z - \frac{x_2^2 - x_1^2}{2} \right\}$$

This likelihood ratio is strictly increasing in z on the support of $\tilde{\pi}(\cdot)$. Hence, since $\tilde{\pi}(\cdot)$ is non-degenerate, the distribution q_{x_2} strictly first-order stochastically dominates q_{x_1} .

Since $1 - \Phi(1.96 - z)$ is strictly increasing in z ,

$$g(x_2) = \mathbb{E}_{q_{x_2}} [1 - \Phi(1.96 - Z)] > \mathbb{E}_{q_{x_1}} [1 - \Phi(1.96 - Z)] = g(x_1)$$

Thus, $g(x)$ is strictly increasing in x .

Since $cv(p) = \Phi^{-1}(1 - p/2)$ is continuous and strictly decreasing in $p \in (0, 1)$, $r(p) = g(cv(p))$ is continuous and strictly decreasing in p . Its image over $p \in (0, 1)$ is therefore $(\underline{r}, \bar{r})$. For any $T \in (\underline{r}, \bar{r})$, the intermediate value theorem implies that there exists a $p^* \in (0, 1)$ such that $r(p^*) = T$. Strict monotonicity implies p^* is unique.

It remains to establish the endpoint results. Since $q_x(z) \propto \exp\left\{xz - \frac{z^2}{2}\right\} \tilde{\pi}(z)$, unbounded support of $\tilde{\pi}(\cdot)$ above implies that q_x shifts arbitrarily far to the right as $x \rightarrow \infty$. Hence $Z \rightarrow \infty$ in probability under q_x as $x \rightarrow \infty$. Since $1 - \Phi(1.96 - z) \rightarrow 1$ as $z \rightarrow \infty$ and is bounded, it follows that $g(x) \rightarrow 1$. Therefore, $\bar{r} = 1$.

Finally, since $cv(p) \rightarrow 0$ as $p \rightarrow 1$,

$$\underline{r} = g(0) = \frac{\int_{\mathbb{R}} [1 - \Phi(1.96 - z)] \varphi(z) \tilde{\pi}(z) dz}{\int_{\mathbb{R}} \varphi(z) \tilde{\pi}(z) dz}$$

Using the symmetry of $\varphi(z)$ and $\tilde{\pi}(z)$,

$$\underline{r} = \frac{\int_0^\infty \{[1 - \Phi(1.96 - z)] + [1 - \Phi(1.96 + z)]\} \varphi(z) \tilde{\pi}(z) dz}{2 \int_0^\infty \varphi(z) \tilde{\pi}(z) dz}$$

For every $z \geq 0$,

$$\frac{[1 - \Phi(1.96 - z)] + [1 - \Phi(1.96 + z)]}{2} \geq 1 - \Phi(1.96)$$

since the left-hand side equals $1 - \Phi(1.96)$ at $z = 0$ and has derivative $\frac{\varphi(1.96-z) - \varphi(1.96+z)}{2} > 0$ for $z > 0$. Since $\tilde{\pi}(\cdot)$ is non-degenerate, the inequality is strict. Hence $\underline{r} > 1 - \Phi(1.96) = 0.025$. This completes the proof. $\square$

B. NONPARAMETRIC RESULTS AND PROOFS

In this section we provide general results for nonparametric estimation of replication probabilities. First we formalize and define some key quantities.

Let $\hat{z} = z + V$ where $V \sim N(0, 1)$ and z has unknown density π . Consider a

replication where sample sizes are multiplied by γ^2 . So $\hat{z}_r = \gamma z + V_r$ where $V_r \sim N(0, 1)$. The random variables (V, V_r, Z) are mutually independent.

The meta-analyst wishes to estimate the replication probability $r(p, \gamma)$:

$$\begin{aligned} r(p, \gamma) &\equiv \mathbb{P} [|\hat{z}_r| > c \text{ and } \text{sign}(\hat{z}_r) = \text{sign}(\hat{z}) \mid |\hat{z}| = cv(p), D = 1] \\ &= \frac{\int_{\mathbb{R}} [1 - \Phi(c - \gamma z)] \varphi(cv(p) - z) \tilde{\pi}(z) dz}{\int_{\mathbb{R}} \varphi(cv(p) - z') \tilde{\pi}(z') dz'} \end{aligned}$$

where φ is the normal density, Φ is the normal CDF, $\tilde{\pi}(z) \equiv \frac{\pi(z) + \pi(-z)}{2}$ is the symmetrized density, and $cv(p)$ is the two-sided critical value associated with p -value p .

The meta-analyst observes a random sample of $|\hat{z}|$ —i.e. they observe only absolute values. Moreover, they only observe published $|\hat{z}|$. Let $D = 1$ indicate that a finding is published, and suppose its publication probability depends on the observed t -ratio through the even selection function s_{θ_0} :

$$\Pr(D = 1 \mid \hat{z}) = s_{\theta_0}(\hat{z}) \quad (10)$$

where the functional form of s_{θ} is known but the parameter vector θ_0 is unknown.

B.1 Identification

Theorem B.1 shows that whenever the publication bias parameter θ_0 is identified then the symmetrized latent density $\tilde{\pi}$ is identified and therefore the target parameter $r(p, \gamma)$ is also identified. Moreover, the fact that only the absolute values of test statistics are observed by the meta-analyst means that π itself is unidentified. For step functions, the identification of θ_0 comes immediately from caliper ratios.

Theorem B.1. *Assume that the density π exists, that publication bias follows Equation (10), that the function family $s_{\theta}(z)$ is known, $\inf_{\theta \in \Theta, z \in \mathbb{R}} s_{\theta}(z) > 0$, $s_{\theta}(z)$ is even in z , and that θ_0 is identified from the distribution of published $|\hat{z}|$. Then:*

- $\pi(z)$ is not identified from the distribution of published $|\hat{z}|$.
- $\tilde{\pi}(z) \equiv \frac{\pi(z) + \pi(-z)}{2}$ is identified from the distribution of published $|\hat{z}|$.

Proof. Non-Identification of π : Let S be a Rademacher random variable independent of z . Consider two variables z, z' . z is positive almost surely and $z' \equiv Sz$. So

z, z' have different distributions but yield the same distribution for $|\hat{z}|$. So π is not identified from the distribution of $|\hat{z}|$.

Identification of $f_{|\hat{z}|}$: Let $f_{|\hat{z}|}$ denote the PDF of $|\hat{z}|$. The researcher observes a random sample of published $|\hat{z}|$. So the conditional density $f_{|\hat{z}||D=1}$ is identified. By Bayes' Rule: $f_{|\hat{z}|}(x) = \frac{f_{|\hat{z}||D=1}(x)\mathbb{E}[s_{\theta_0}(\hat{z})]}{s_{\theta_0}(x)}$. By Bayes' rule: $\mathbb{E}[s_{\theta_0}(\hat{z})] = 1/\mathbb{E}[1/s_{\theta_0}(\hat{z})|D=1]$. So $f_{|\hat{z}|}(x) = \frac{f_{|\hat{z}||D=1}(x)}{s_{\theta_0}(x)\mathbb{E}[1/s_{\theta_0}(\hat{z})|D=1]}$. By assumption the function family s_{θ} is known and even in z . So if θ_0 is identified then the unconditional density $f_{|\hat{z}|}$ is also identified.

Identification of $\tilde{f}_{\hat{z}}$: Define $\tilde{f}_{\hat{z}}(x) \equiv \frac{f_{\hat{z}}(x) + f_{\hat{z}}(-x)}{2}$. This is the symmetrized version of the PDF of $\hat{z}$. Since $\tilde{f}_{\hat{z}}(x) \equiv f_{|\hat{z}|}(|x|)/2$ and we already showed that $f_{|\hat{z}|}$ is identified, then $\tilde{f}_{\hat{z}}$ is also identified.

Identification of $\tilde{\pi}$: Let S be a Rademacher random variable independent of $(z, V, V_r, \hat{z})$. Then $S|\hat{z}|$ and $S\hat{z}$ both share the same PDF $\tilde{f}_{\hat{z}}$. Since we already showed that $\tilde{f}_{\hat{z}}$ is identified, the distribution of $S\hat{z}$ is identified. Notice that $S\hat{z} = Sz + SV \stackrel{d}{=} Sz + V$. Since its density is identified, the characteristic function $\phi_{S\hat{z}}(\zeta)$ is identified. By the deconvolution theorem, $\phi_{Sz}(\zeta)\exp(-\zeta^2/2) = \phi_{S\hat{z}}(\zeta)$ where ϕ_{Sz} is the characteristic function of Sz and $\phi_{S\hat{z}}$ is the characteristic function of $S\hat{z}$. Since $\exp(-\zeta^2/2) > 0$ for all ζ , the distribution of Sz is identified. Since Sz has PDF $\tilde{\pi}$, so $\tilde{\pi}$ is identified. □

Since the replication probability $r(p, \gamma)$ is simply a known integral over $\tilde{\pi}$ we have the corollary:

Corollary 1. *Let the assumptions of Theorem B.1 hold. Then, for all $\gamma > 0, p \in (0, 1)$, the replication probability $r(p, \gamma)$ is identified from the distribution of published $|\hat{z}|$.*

B.2 Estimation

Define $\mu_j \equiv 2^{-j/2}b_j$. For any $\gamma > 0$, the meta-analyst uses an estimator of the same form as before:

$$\begin{aligned}\hat{r}^{(np)}(p, \gamma) &\equiv 1 - \frac{\sum_{j=0}^{J_n} a_j 2^{j/2} \hat{\mu}_j}{\sum_{j=0}^{J_n} d_j 2^{j/2} \hat{\mu}_j} \\ \hat{\mu}_j &\equiv \frac{1}{n} \sum_{i=1}^n \frac{\psi_j(cv(p) - |\hat{z}_i|) \varphi(cv(p) - |\hat{z}_i|) + \psi_j(cv(p) + |\hat{z}_i|) \varphi(cv(p) + |\hat{z}_i|)}{s_{\hat{\theta}_n}(\hat{z}_i)}\end{aligned}$$

But, the known deterministic scalar weights a_j, d_j are now the known deterministic constants:

$$\begin{aligned}a_j &\equiv \int_{\mathbb{R}} \varphi(u) \exp(-u^2/2) \Phi(c - \gamma cv(p) + \sqrt{2}\gamma u) \psi_j(u) du \\ d_j &\equiv \int_{\mathbb{R}} \varphi(u) \exp(-u^2/2) \psi_j(u) du\end{aligned}$$

Theorem B.2 finds the rate of consistency of $\hat{r}^{(np)}(p, \gamma)$ and shows that this rate slows down as γ gets larger.

Theorem B.2. *Let the sample $|\hat{z}_1|, \dots, |\hat{z}_n|$ satisfy Assumption 1. Assume that the density π exists and has bounded height and that publication bias follows Equation (10). Assume that the researcher has an estimator $\hat{\theta}_n$ of the publication bias parameters θ_0 such that for some constant $q \in (0, \frac{1}{2}]$:*

$$\sup_{t \in \mathbb{R}} \left| \frac{1}{s_{\hat{\theta}_n}(t)} - \frac{1}{s_{\theta_0}(t)} \right| = \mathcal{O}_p(n^{-q})$$

Then for fixed p, c, γ if the researcher sets $J_n = \lceil \frac{2q}{\log 2} \log(An) \rceil$ for any $A > 0$, then:

$$\hat{r}^{(np)}(p, \gamma) - r(p, \gamma) = \mathcal{O}_p \left(n^{q \log_2 \left(\frac{1+2\gamma^2}{2+2\gamma^2} \right)} (\log n)^{3/4} \right)$$

If instead θ_0 is known, then:

$$\hat{r}^{(np)}(p, \gamma) - r(p, \gamma) = \mathcal{O}_p \left(n^{\frac{1}{2} \log_2 \left(\frac{1+2\gamma^2}{2+2\gamma^2} \right)} (\log n)^{3/4} \right)$$

Proof. Define the inner products:

$$\begin{aligned}\langle g_1, g_2 \rangle_T &\equiv \int_{-\infty}^{\infty} g_1(x) g_2(x) \varphi(x) dx \\ \langle g_1, g_2 \rangle_Z &\equiv \int_{-\infty}^{\infty} g_1(x) g_2(x) \varphi(x/\sqrt{2})/\sqrt{2} dx\end{aligned}$$

Recall that if S is an independent Rademacher, then $S|\hat{z}| \stackrel{d}{=} S\hat{z} \stackrel{d}{=} Sz + V$ where $V \sim N(0, 1)$ and $V \perp Sz$. Sz has PDF $\tilde{\pi}$ and $S|\hat{z}|, S\hat{z}$ share the PDF $\tilde{f}_{\hat{z}}$ which is the symmetrized version of the PDF of $\hat{z}$.

Since $\sup_j |\langle \chi_j, \tilde{\pi} \rangle_Z| < \infty$ by [Faridani \(2026\)](#), $|\langle \psi_j, \tilde{f}_{\hat{z}} \rangle_T| = \mathcal{O}(2^{-j/2})$.

Substituting $z = cv(p) - u$ we have:

$$\begin{aligned}r(p, \gamma) &= \frac{\int_{\mathbb{R}} [1 - \Phi(c - \gamma z)] \varphi(cv(p) - z) \tilde{\pi}(z) dz}{\int_{\mathbb{R}} \varphi(cv(p) - z') \tilde{\pi}(z') dz'} \\ &= 1 - \frac{\int_{\mathbb{R}} \varphi(cv(p) - z) \tilde{\pi}(z) \Phi(c - \gamma z) dz}{\int_{\mathbb{R}} \varphi(cv(p) - z') \tilde{\pi}(z') dz'} \\ &= 1 - \frac{\int_{\mathbb{R}} \varphi(u) \tilde{\pi}(cv(p) - u) \Phi(c - \gamma cv(p) + \gamma u) du}{\int_{\mathbb{R}} \varphi(u') \tilde{\pi}(cv(p) - u') du'} \\ &= 1 - \frac{\int_{\mathbb{R}} \varphi(u) \tilde{\pi}_p(u) \Phi(c - \gamma cv(p) + \gamma u) du}{\int_{\mathbb{R}} \varphi(u') \tilde{\pi}_p(u') du'} \\ &= 1 - \frac{\int_{\mathbb{R}} \varphi(u/\sqrt{2}) \tilde{\pi}_p(u) \exp(-u^2/4) \Phi(c - \gamma cv(p) + \gamma u) du}{\int_{\mathbb{R}} \varphi(u/\sqrt{2}) \exp(-u^2/4) \tilde{\pi}_p(u) du} \\ &= 1 - \frac{\langle \tilde{\pi}_p, \exp(-u^2/4) \Phi(c - \gamma cv(p) + \gamma u) \rangle_Z}{\langle \tilde{\pi}_p, \exp(-u^2/4) \rangle_Z}\end{aligned}$$

where $\tilde{\pi}_p(z) \equiv \tilde{\pi}(cv(p) - z)$.

Define $\chi_j(x) \equiv \frac{\text{He}_j(x/\sqrt{2})}{\sqrt{j!}}$. Then since the χ_j form an orthonormal basis:

$$\begin{aligned}\exp(-u^2/4) \Phi(c - \gamma cv(p) + \gamma u) &= \sum_{j=0}^{\infty} a_j \chi_j(u) \\ \exp(-u^2/4) &= \sum_{j=0}^{\infty} d_j \chi_j(u) \\ a_j &= \langle \exp(-u^2/4) \Phi(c - \gamma cv(p) + \gamma u), \chi_j \rangle_Z \\ d_j &= \langle \exp(-u^2/4), \chi_j \rangle_Z\end{aligned}$$

The χ_j form an orthonormal basis of the set of all functions with finite norm $\langle f, f \rangle_Z < \infty$ and the assumed bounded height of π guarantees that $\langle \tilde{\pi}_p, \tilde{\pi}_p \rangle_Z < \infty$. So by Parseval's Formula there are unknown coefficients b_j such that:

$$\begin{aligned}\tilde{\pi}_p(z) &= \sum_{j=0}^{\infty} b_j \chi_j(z) \\ b_j &= \langle \tilde{\pi}_p, \chi_j \rangle_Z = \mathbb{E} \left[\varphi((cv(p) - S|z|)/\sqrt{2}) \chi_j(cv(p) - S|z|/\sqrt{2}) \right]\end{aligned}$$

By the orthonormality of χ_j and the cancelling of common factors:

$$r(p, \gamma) = 1 - \frac{\sum_{j=0}^{\infty} a_j b_j}{\sum_{j=0}^{\infty} d_j b_j}$$

Again by the orthonormality of the Hermite polynomials:

$$\begin{aligned}a_j &= \left\langle \exp(-u^2/4) \Phi(c - \gamma cv(p) + \gamma u), \frac{\text{He}_j(u/\sqrt{2})}{\sqrt{j!}} \right\rangle_Z \\ d_j &= \left\langle \exp(-u^2/4), \frac{\text{He}_j(u/\sqrt{2})}{\sqrt{j!}} \right\rangle_Z\end{aligned}$$

Thus the bias of approximating $r(p, \gamma)$ with a sum up to J_n decays exponentially fast in J_n :

$$\begin{aligned}r(p, \gamma) - \left(1 - \frac{\sum_{j=0}^{J_n} a_j b_j}{\sum_{j=0}^{J_n} d_j b_j} \right) &= \frac{\sum_{j=0}^{\infty} a_j b_j}{\sum_{j=0}^{\infty} d_j b_j} - \frac{\sum_{j=0}^{J_n} a_j b_j}{\sum_{j=0}^{J_n} d_j b_j} \\ &\equiv \frac{N_0}{D_0} - \frac{N_n}{D_n} \\ &= \frac{N_0 D_n - N_n D_0}{D_0 D_n} \\ &= \frac{N_n D_n + (N_0 - N_n) D_n - N_n D_n - N_n (D_0 - D_n)}{D_0 D_n} \\ &= \frac{(N_0 - N_n) D_n - N_n (D_0 - D_n)}{D_0 D_n}\end{aligned}$$

Since $D_0 > 0$ and $D_n \rightarrow D_0$, then for large enough J_n , $D_n > D_0/2 > 0$. Thus,

using Lemmas B.2 and B.3 and the fact that the $|b_j|$ are bounded because $\langle \tilde{\pi}, \tilde{\pi} \rangle_Z < \infty$:

$$\begin{aligned}
\left| \frac{(N_0 - N_n)D_n - N_n(D_0 - D_n)}{D_0 D_n} \right| &\lesssim |N_0 - N_n| + |D_0 - D_n| \\
\left| r(p, \gamma) - \left(1 - \frac{\sum_{j=0}^{J_n} a_j b_j}{\sum_{j=0}^{J_n} d_j b_j} \right) \right| &\lesssim \left| \sum_{j=J_n+1}^{\infty} a_j b_j \right| + \left| \sum_{j=J_n+1}^{\infty} d_j b_j \right| \\
&\lesssim \sum_{j=J_n+1}^{\infty} |a_j| + \sum_{j=J_n+1}^{\infty} |d_j| \\
&\lesssim \sum_{j=J_n+1}^{\infty} j^{3/4} \left(\frac{1 + 2\gamma^2}{2 + 2\gamma^2} \right)^{j/2} + \frac{1}{2^{j/2}}
\end{aligned}$$

The dominant truncation bias comes from the numerator: $\sum_{j=J_n+1}^{\infty} j^{3/4} \left(\frac{1+2\gamma^2}{2+2\gamma^2} \right)^{j/2}$.

For ease of notation let $\rho_r \equiv \sqrt{\frac{1+2\gamma^2}{2+2\gamma^2}}$

$$\begin{aligned}
\sum_{j=J_n+1}^{\infty} j^{3/4} \left(\frac{1 + 2\gamma^2}{2 + 2\gamma^2} \right)^{j/2} &= \sum_{j=J_n+1}^{\infty} j^{3/4} \rho_r^j \\
&= \rho_r^{J_n+1} (J_n + 1)^{3/4} \sum_{k=0}^{\infty} \left(1 + \frac{k}{J_n + 1} \right)^{3/4} \rho_r^k \\
&\leq \rho_r^{J_n+1} (J_n + 1)^{3/4} \sum_{k=0}^{\infty} 2(1 + k) \rho_r^k \\
&= \rho_r^{J_n+1} (J_n + 1)^{3/4} \frac{2}{(1 - \rho_r)^2} \\
&= \mathcal{O}(\rho_r^{J_n} J_n^{3/4}) = \mathcal{O} \left(\left(\frac{1 + 2\gamma^2}{2 + 2\gamma^2} \right)^{J_n/2} J_n^{3/4} \right)
\end{aligned}$$

So the bias decays with:

$$\left| r(p, \gamma) - \left(1 - \frac{\sum_{j=0}^{J_n} a_j b_j}{\sum_{j=0}^{J_n} d_j b_j} \right) \right| \lesssim \left(\frac{1 + 2\gamma^2}{2 + 2\gamma^2} \right)^{J_n/2} J_n^{3/4}$$

The a_j, d_j are known and can be computed via numerical integration but the b_j must be estimated. To do the estimation first notice that:

$$b_j = \langle \tilde{\pi}_p, \chi_j \rangle_Z = \mathbb{E} \left[\chi_j (cv(p) - Sz) \varphi((cv(p) - Sz)/\sqrt{2})/\sqrt{2} \right]$$

Since the $\sqrt{2}$ appears in both numerator and denominator of $r(p, \gamma)$, it can be ignored from now on. By the Singular Value Decomposition in Example 1 of [Carrasco and Florens \(2011\)](#) and [Faridani \(2026\)](#):

$$\begin{aligned} \mathbb{E} \left[\chi_j(cv(p) - SZ) \varphi((cv(p) - SZ)/\sqrt{2}) \right] &= 2^{j/2} \mathbb{E} \left[\psi_j(cv(p) - S\hat{Z}) \varphi(cv(p) - S\hat{Z}) \right] \\ &= 2^{j/2} \mathbb{E} \left[\psi_j(cv(p) - S|\hat{Z}|) \varphi(cv(p) - S|\hat{Z}|) \right] \\ &= \frac{1}{2} 2^{j/2} \mathbb{E} \left[\psi_j(cv(p) - |\hat{Z}|) \varphi(cv(p) - |\hat{Z}|) \right] \\ &\quad + \frac{1}{2} 2^{j/2} \mathbb{E} \left[\psi_j(cv(p) + |\hat{Z}|) \varphi(cv(p) + |\hat{Z}|) \right] \end{aligned}$$

Substituting and multiplying numerator and denominator by a constant:

$$\begin{aligned} r(p, \gamma) &= 1 - \frac{\sum_{j=0}^{\infty} a_j 2^{j/2} \mu_j}{\sum_{j=0}^{\infty} d_j 2^{j/2} \mu_j} \\ \mu_j &\equiv \mathbb{E} \left[\psi_j(cv(p) - |\hat{Z}|) \varphi(cv(p) - |\hat{Z}|) \right] \frac{1}{\mathbb{P}(D=1)} \\ &\quad + \mathbb{E} \left[\psi_j(cv(p) + |\hat{Z}|) \varphi(cv(p) + |\hat{Z}|) \right] \frac{1}{\mathbb{P}(D=1)} \end{aligned}$$

Therefore, our estimator of $r(p, \gamma)$ is:

$$\hat{r}(p, \gamma) \equiv 1 - \frac{\sum_{j=0}^{J_n} a_j 2^{j/2} \hat{\mu}_j}{\sum_{j=0}^{J_n} d_j 2^{j/2} \hat{\mu}_j}$$

By Lemma [B.1](#), for any non-negative weights α_j the estimators $\hat{\mu}_j$ satisfy:

$$\sum_{j=0}^{J_n} \alpha_j |\hat{\mu}_j - \mu_j| \lesssim n^{-q} \sum_{j=0}^{J_n} |\alpha_j|$$

So we can bound the stochastic part of the estimation error.

$$\begin{aligned}
b_j &= \frac{\mathbb{P}(D=1)}{2\sqrt{2}} 2^{j/2} \mu_j \\
\sum_{j=0}^{J_n} a_j 2^{j/2} \hat{\mu}_j - \frac{2\sqrt{2}}{\mathbb{P}(D=1)} \sum_{j=0}^{J_n} a_j b_j &= \sum_{j=0}^{J_n} a_j 2^{j/2} \hat{\mu}_j - \sum_{j=0}^{J_n} a_j 2^{j/2} \mu_j \\
&= \mathcal{O}_p \left(n^{-q} \sum_{j=0}^{J_n} |a_j| 2^{j/2} \right) \\
&= \mathcal{O}_p \left(n^{-q} \sum_{j=0}^{J_n} j^{3/4} 2^{j/2} \left(\frac{1+2\gamma^2}{2+2\gamma^2} \right)^{j/2} \right) \\
&= \mathcal{O}_p \left(n^{-q} J_n^{3/4} 2^{J_n/2} \left(\frac{1+2\gamma^2}{2+2\gamma^2} \right)^{J_n/2} \right) \\
\sum_{j=0}^{J_n} d_j 2^{j/2} \hat{\mu}_j - \frac{2\sqrt{2}}{\mathbb{P}(D=1)} \sum_{j=0}^{J_n} d_j b_j &= \mathcal{O}_p \left(n^{-q} \sum_{j=0}^{J_n} 2^{j/2} |d_j| \right) = \mathcal{O}_p(n^{-q} J_n)
\end{aligned}$$

We already showed that for large enough J_n , $\left| \sum_{j=0}^{J_n} d_j 2^{j/2} \mu_j \right| > \left| \sum_{j=0}^{\infty} d_j 2^{j/2} \mu_j \right| / 2$. So with high probability the denominator is bounded away from zero:

$$\mathbb{P} \left(\left| \sum_{j=0}^{J_n} d_j 2^{j/2} \hat{\mu}_j \right| > \left| \sum_{j=0}^{\infty} d_j 2^{j/2} \mu_j \right| / 4 \right) \rightarrow 1$$

The factors of $\frac{2\sqrt{2}}{\mathbb{P}(D=1)}$ cancel in the fraction. Notice also that the stochastic error of the denominator is dominated. Thus:

$$\hat{r}(p, \gamma) - \left(1 - \frac{\sum_{j=0}^{J_n} a_j b_j}{\sum_{j=0}^{J_n} d_j b_j} \right) = \mathcal{O}_p \left(n^{-q} J_n + n^{-q} J_n^{3/4} 2^{J_n/2} \left(\frac{1+2\gamma^2}{2+2\gamma^2} \right)^{J_n/2} \right)$$

For fixed γ , the $n^{-q} J_n$ term is dominated. We already bounded the bias. Thus the total estimation error is:

$$\hat{r}(p, \gamma) - r(p, \gamma) = \mathcal{O}_p \left(\left(\frac{1+2\gamma^2}{2+2\gamma^2} \right)^{J_n/2} J_n^{3/4} (n^{-q} 2^{J_n/2} + 1) \right)$$

For any $A > 0$, suppose we set:

$$J_n = \lceil \beta \log(An) \rceil$$

Then the $J_n^{3/4}$ factor in the estimation error will only affect the rate up to log factors so we ignore it in the optimization. Now we optimize over $\beta > 0$:

If $n^{-q}2^{J_n/2} \lesssim 1$, then $2^{J_n/2} = \mathcal{O}(n^q)$, so $J_n \sim 2 \log_2(Dn^q) = \frac{2q}{\log 2} \log(An)$. So the final rate is:

$$\begin{aligned} \hat{r}(p, \gamma) - r(p, \gamma) &= \mathcal{O}_p \left(\left(\frac{1 + 2\gamma^2}{2 + 2\gamma^2} \right)^{\frac{q}{\log 2} \log(n)} (\log n)^{3/4} \right) \\ &= \mathcal{O}_p \left(n^{q \log_2 \left(\frac{1 + 2\gamma^2}{2 + 2\gamma^2} \right)} (\log n)^{3/4} \right) \end{aligned}$$

□

Lemma B.1. *Let $q \leq \frac{1}{2}$. Let the assumptions of Theorem B.1 hold. For any weights α_j the estimators $\hat{\mu}_j$ satisfy:*

$$\sum_{j=0}^{J_n} |\alpha_j| |\hat{\mu}_j - \mu_j| = \mathcal{O}_p \left(n^{-q} \sum_{j=0}^{J_n} |\alpha_j| \right)$$

Proof.

$$\mu_j \equiv \mathbb{E} \left[\psi_j(cv(p) - |\hat{Z}|) \varphi(cv(p) - |\hat{Z}|) \right] \frac{1}{\mathbb{P}(D = 1)} \quad (11)$$

$$+ \mathbb{E} \left[\psi_j(cv(p) + |\hat{Z}|) \varphi(cv(p) + |\hat{Z}|) \right] \frac{1}{\mathbb{P}(D = 1)} \quad (12)$$

Define the shorthand $v(x)$:

$$\begin{aligned} v_j(x) &= \psi_j(cv(p) - |x|) \varphi(cv(p) - |x|) + \psi_j(cv(p) + |x|) \varphi(cv(p) + |x|) \\ \mu_j &= \mathbb{E} [v_j(|\hat{Z}|)] \frac{1}{\mathbb{P}(D = 1)} \end{aligned}$$

Now add publication bias. Recall that the conditional probability of reporting is equal to $\Pr(D = 1 \mid T = t) = s_{\theta_0}(t)$. In the presence of publication bias, the $\hat{z}_i$ need

to be weighted to remove it using Bayes' Rule:

$$\mathbb{E}[v_j(|\hat{z}|)] \frac{1}{\mathbb{P}(D=1)} = \mathbb{E}\left[\frac{v_j(|\hat{z}|)}{s_{\theta_0}(\hat{z})} \mid D=1\right]$$

Since s is even in $\hat{z}$ by assumption, $s_{\theta}(\hat{z}) = s_{\theta}(|\hat{z}|)$. Thus:

$$\begin{aligned}\mu_j &= \mathbb{E}\left[\frac{v_j(|\hat{z}|)}{s_{\theta_0}(\hat{z})} \mid D=1\right] \\ \hat{\mu}_j &= \frac{1}{n} \sum_{i=1}^n \frac{v_j(|\hat{z}_i|)}{s_{\hat{\theta}_n}(|\hat{z}_i|)}\end{aligned}$$

Suppose we have an estimator $\hat{\theta}_n$ such that for some fixed $q \in (0, \frac{1}{2}]$:

$$\sup_{t \in \mathbb{R}} \left| \frac{1}{s_{\hat{\theta}_n}(t)} - \frac{1}{s_{\theta_0}(t)} \right| = \mathcal{O}_p(n^{-q})$$

Define $\tilde{\mu}_j \equiv \frac{1}{n} \sum_{i=1}^n \frac{v_j(|\hat{z}_i|)}{s_{\theta_0}(|\hat{z}_i|)}$. Now we break the estimation error into two pieces.

$$\sum_{j=0}^{J_n} \alpha_j (\hat{\mu}_j - \mu_j) = \sum_{j=0}^{J_n} \alpha_j (\tilde{\mu}_j - \mu_j) + \sum_{j=0}^{J_n} \alpha_j (\hat{\mu}_j - \tilde{\mu}_j) \quad (13)$$

Now we bound the first sum of (13). Notice that since $|\hat{z}_i|$ are identically distributed, $\mathbb{E}[\tilde{\mu}_j] = \mu_j$. By assumption, $\inf_{t, \theta} s_{\theta}(t) > 0$ and by [Indritz \(1961\)](#) we have $\sup_{j,x} |\varphi(x)\psi_j(x)| < \infty$. So $\sup_{j,x} |v_j(x)| < \infty$. Moreover, since observations are assumed to be clustered each summand in $\tilde{\mu}_j$ is correlated with at most $L \geq 1$ others. So there is some constant $C > 0$ such that:

$$\begin{aligned}\sup_{j \in \{1,2,\dots\}} \mathbb{E}[|\mu_j - \tilde{\mu}_j|] &\leq \sup_{j \in \{1,2,\dots\}} \sqrt{\mathbb{V}[\tilde{\mu}_j]} \leq \frac{C}{\sqrt{n}} \\ \sum_{j=0}^{J_n} |\alpha_j| \mathbb{E}[|\mu_j - \tilde{\mu}_j|] &\leq \frac{C}{\sqrt{n}} \sum_{j=0}^{J_n} |\alpha_j|\end{aligned}$$

All moments are implicitly conditional on publication of the sample of $\hat{z}_i$. By

Markov's Inequality:

$$\sum_{j=0}^{J_n} |\alpha_j| |\tilde{\mu}_j - \mu_j| = \mathcal{O}_p \left(n^{-1/2} \sum_{j=0}^{J_n} |\alpha_j| \right)$$

Now we bound the second sum of (13). By assumption, $\inf_{t,\theta} s_\theta(t) > 0$ for all θ, t . Therefore:

$$\begin{aligned} \sup_{j \in \{1, \dots, J_n\}} |\hat{\mu}_j - \tilde{\mu}_j| &= \mathcal{O}_p(n^{-q}) \\ \sum_{j=0}^{J_n} |\alpha_j| |\hat{\mu}_j - \tilde{\mu}_j| &= \mathcal{O}_p \left(n^{-q} \sum_{j=0}^{J_n} |\alpha_j| \right) \end{aligned}$$

Since $q \leq \frac{1}{2}$, the second sum is the dominant one in (13):

$$\sum_{j=0}^{J_n} |\alpha_j| |\hat{\mu}_j - \mu_j| = \mathcal{O}_p \left(n^{-q} \sum_{j=0}^{J_n} |\alpha_j| \right)$$

□

Lemma B.2. *Let $\gamma > 0$ be the square root of the ratio of the replication sample size to the original sample size. Then: $|a_j| \lesssim j^{3/4} \left(\frac{1+2\gamma^2}{2+2\gamma^2} \right)^{j/2}$ for $j > 0$.*

Proof.

$$\begin{aligned} a_j &= \int_{\mathbb{R}} \varphi(u/\sqrt{2})/\sqrt{2} \exp(-u^2/4) \Phi(c - \gamma cv(p) + \gamma u) \chi_j(u) du \\ &= \int_{\mathbb{R}} \varphi(u) \exp(-u^2/4) \Phi(c - \gamma cv(p) + \sqrt{2}\gamma u) \chi_j(u\sqrt{2}) du \\ &= \int_{\mathbb{R}} \varphi(u) \exp(-u^2/2) \Phi(c - \gamma cv(p) + \sqrt{2}\gamma u) \frac{\text{He}_j(u)}{\sqrt{j!}} du \end{aligned}$$

The generating function of the Hermite polynomials (DLMF 18.12.16) is:

$$\sum_{j=0}^{\infty} \text{He}_j(x) \frac{t^j}{j!} = \exp \left(xt - \frac{t^2}{2} \right)$$

Substituting the Hermite generating function in:

$$\begin{aligned}
F(t) &\equiv \sum_{j=0}^{\infty} a_j \frac{t^j}{\sqrt{j!}} \\
&= \int_{\mathbb{R}} \varphi(u) \exp(-u^2/2) \Phi(c - \gamma cv(p) + \sqrt{2}\gamma u) \left(\sum_{j=0}^{\infty} \frac{\text{He}_j(u)}{j!} \frac{t^j}{\sqrt{j!}} \right) du \\
&= \int_{\mathbb{R}} \varphi(u) \exp(-u^2/2) \Phi(c - \gamma cv(p) + \sqrt{2}\gamma u) \exp\left(ut - \frac{t^2}{2}\right) du \\
&= \frac{1}{\sqrt{2\pi}} \int_{\mathbb{R}} \Phi(c - \gamma cv(p) + \sqrt{2}\gamma u) \exp\left(ut - \frac{t^2}{2} - u^2\right) du
\end{aligned}$$

By completing the square: $ut - \frac{t^2}{2} - u^2 = -\left(u - \frac{t}{2}\right)^2 - \frac{t^2}{4}$. Thus:

$$F(t) = \frac{\exp(-t^2/4)}{\sqrt{2\pi}} \int_{\mathbb{R}} \Phi(c - \gamma cv(p) + \sqrt{2}\gamma u) \exp\left(-\left(u - \frac{t}{2}\right)^2\right) du$$

First consider the case for real t . Let the random variable U be distributed: $U \sim N(\frac{t}{2}, \frac{1}{2})$. This means that $\sqrt{2}\gamma U \sim N(\frac{\gamma}{\sqrt{2}}t, \gamma^2)$. Thus:

$$\begin{aligned}
F(t) &= \frac{\exp(-t^2/4)}{\sqrt{2}} \mathbb{E} \left[\Phi(c - \gamma cv(p) + \sqrt{2}\gamma U) \right] \\
&= \frac{\exp(-t^2/4)}{\sqrt{2}} \mathbb{P} \left[Z \leq c - \gamma cv(p) + \sqrt{2}\gamma U \right] \\
&= \frac{\exp(-t^2/4)}{\sqrt{2}} \mathbb{P} \left[Z_1 \leq c - \gamma cv(p) + \gamma Z_2 + \frac{\gamma t}{\sqrt{2}} \right] \\
&= \frac{\exp(-t^2/4)}{\sqrt{2}} \mathbb{P} \left[(Z_1 - \gamma Z_2)/\sqrt{1 + \gamma^2} \leq (c - \gamma cv(p))/\sqrt{1 + \gamma^2} + \frac{t\gamma}{\sqrt{2(1 + \gamma^2)}} \right] \\
&= \frac{\exp(-t^2/4)}{\sqrt{2}} \Phi \left((c - \gamma cv(p))/\sqrt{1 + \gamma^2} + \frac{t\gamma}{\sqrt{2(1 + \gamma^2)}} \right)
\end{aligned}$$

Since both left and right hand sides are entire functions, the equation also holds for complex t . So we have a generating function for the coefficients a_j :

$$F(t) \equiv \sum_{j=0}^{\infty} a_j \frac{t^j}{\sqrt{j!}} = \frac{\exp(-t^2/4)}{\sqrt{2}} \Phi \left((c - \gamma cv(p))/\sqrt{1 + \gamma^2} + \frac{t\gamma}{\sqrt{2(1 + \gamma^2)}} \right)$$

This means that the a_j are equal to the derivatives of $F(t)$ evaluated at zero:

$$a_j = \frac{F^{(j)}(0)}{\sqrt{j!}}$$

So to bound a_j we need only bound the derivatives of F . Since the exponential and the normal CDF are entire functions, F is entire. So we can use the Cauchy Differentiation Formula (DLMF 3.4.17) where we integrate around any circle of radius R in the complex plane centered at zero:

$$F^{(j)}(0) = \frac{j!}{2\pi i} \int_{|t|=R} \frac{F(t)}{t^{j+1}} dt$$

This can be bounded in absolute value by taking the modulus of both sides:

$$|F^{(j)}(0)| \leq j! \frac{1}{R^j} \sup_{|t|=R} |F(t)|$$

Since F is entire, any R is allowed so we can set $R = \sqrt{j}$. Next we bound $|F(t)|$ in the complex plane. The exponential component can be bounded by $\exp(R^2/4)$. The normal CDF can be bounded in the following way. For any $a \in \mathbb{R}$ and $\alpha > 0$:

$$\begin{aligned} \Phi(x) &= \frac{1}{2} \left(1 + \operatorname{erf}(x/\sqrt{2}) \right) \\ \operatorname{erf}\left(x/\sqrt{2}\right) &= \frac{2}{\sqrt{\pi}} \int_0^{x/\sqrt{2}} \exp(-t^2) dt \\ |\operatorname{erf}\left(x/\sqrt{2}\right)| &\leq |x| \exp(\operatorname{Im}(x)^2/2) \\ |\Phi(x)| &\leq 1 + |x| \exp(\operatorname{Im}(x)^2/2) \\ \sup_{|t|=R} |\Phi(a + t\alpha)| &\lesssim 1 + (|a| + \alpha R) \exp(R^2\alpha^2/2) \end{aligned}$$

Let $a = (c - \gamma cv(p))/\sqrt{1 + \gamma^2}$ and $\alpha = \frac{\gamma}{\sqrt{2(1+\gamma^2)}}$. Substituting into F :

$$\begin{aligned} \sup_{|t|=R} |F(t)| &\lesssim \sup_{|t|=R} \frac{1}{\sqrt{2}} \left| \exp(-t^2/4) \Phi(a + \alpha t) \right| \\ &\lesssim \exp(R^2/4) (1 + (|a| + \alpha R) \exp(R^2\alpha^2/2)) \end{aligned}$$

Notice that $\alpha \in (0, 1)$. For $R > \max\{|a|/(1 - \alpha), 2/(1 - \alpha)\}$, we have $|a| + \alpha R \leq R$

and:

$$\begin{aligned}
\sup_{|t|=R} |F(t)| &\lesssim \exp(R^2/4) R \exp(R^2 \alpha^2/2) \\
&= R \exp(R^2(1/4 + \alpha^2/2)) \\
&= R \exp\left(R^2/4 \frac{1 + \gamma^2 + \gamma^2}{(1 + \gamma^2)}\right) \\
&= R \exp\left(R^2 \frac{1 + 2\gamma^2}{4(1 + \gamma^2)}\right)
\end{aligned}$$

Substituting into Cauchy's Differentiation Formula:

$$\begin{aligned}
|F^{(j)}(0)| &\leq j! \frac{1}{R^j} \sup_{|t|=R} |F(t)| \\
&\lesssim j! \frac{1}{R^j} R \exp\left(R^2 \frac{1 + 2\gamma^2}{4(1 + \gamma^2)}\right) \\
|a_j| &\lesssim \sqrt{j!} \frac{1}{R^j} R \exp\left(R^2 \frac{1 + 2\gamma^2}{4(1 + \gamma^2)}\right)
\end{aligned}$$

Stirling's approximation says that $\sqrt{j!} \sim \left(\frac{j}{e}\right)^{j/2} (2\pi j)^{1/4}$. So:

$$\begin{aligned}
|a_j| &\lesssim \left(\frac{j}{e}\right)^{j/2} j^{1/4} \frac{1}{R^j} R \exp\left(R^2 \frac{1 + 2\gamma^2}{4(1 + \gamma^2)}\right) \\
&= j^{j/2} j^{1/4} \frac{1}{R^j} R \exp\left(R^2 \frac{1 + 2\gamma^2}{4(1 + \gamma^2)} - \frac{j}{2}\right)
\end{aligned}$$

Setting $R = \sqrt{j} \frac{\sqrt{2+2\gamma^2}}{\sqrt{1+2\gamma^2}}$ (which is allowed because F is entire and eventually exceeds the constant $\max\{|a|/(1 - \alpha), 2/(1 - \alpha)\}$):

$$\begin{aligned}
|a_j| &\lesssim j^{j/2} j^{1/4} \sqrt{j} j^{-j/2} \left(\frac{\sqrt{1 + 2\gamma^2}}{\sqrt{2 + 2\gamma^2}}\right)^j \\
&= j^{3/4} \left(\frac{1 + 2\gamma^2}{2 + 2\gamma^2}\right)^{j/2}
\end{aligned}$$

□

Lemma B.3. $|d_j| \lesssim 2^{-j/2}$ for $j > 0$

Proof. By an identical argument to the proof of Lemma B.2

$$G(t) = \sum_{j=0}^{\infty} d_j \frac{t^j}{\sqrt{j!}} = \exp(-t^2/4) / \sqrt{2}$$

$$d_j = \frac{G^{(j)}(0)}{\sqrt{j!}}$$

Using the Taylor Expansion:

$$\exp(-t^2/4) = \sum_{k=0}^{\infty} \frac{(-1)^k}{4^k k!} t^{2k}$$

So

$$G^{(2k)}(0) = \frac{(-1)^k (2k)!}{\sqrt{2} 4^k k!}$$

$$d_{2k} = \frac{(-1)^k \sqrt{(2k)!}}{\sqrt{2} 4^k k!}$$

So for odd j , $d_j = 0$. For even coefficients, notice that the ratio of successive terms is:

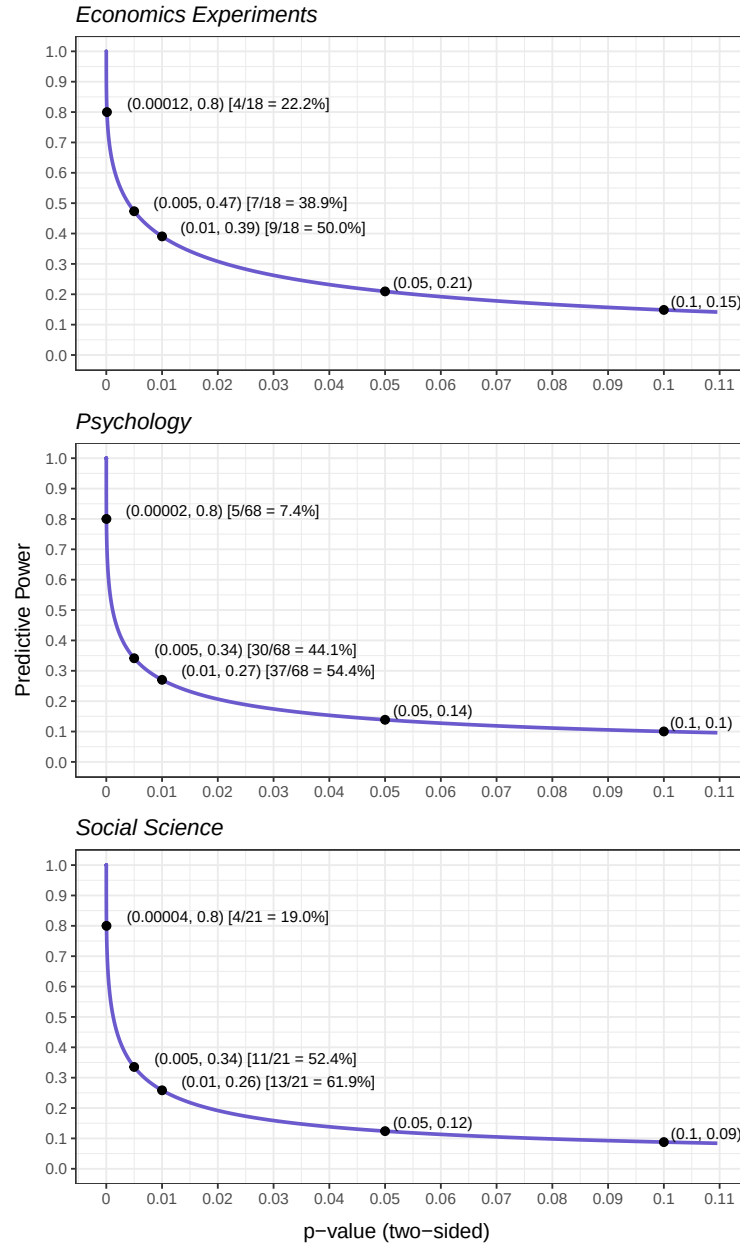
$$\begin{aligned} \left| \frac{d_{2(k+1)}}{d_{2k}} \right| &= \sqrt{\frac{(2k+2)!}{(2k)!} \frac{4^k k!}{4^{k+1} (k+1)!}} \\ &= \frac{\sqrt{(2k+2)(2k+1)}}{4(k+1)} \\ &= \frac{\sqrt{(2k+2)(2k+1)}}{2(2k+2)} \\ &= \sqrt{\frac{(2k+1)}{4(2k+2)}} \\ &= \frac{1}{2} \sqrt{\frac{(2k+1)}{(2k+2)}} \\ &\leq \frac{1}{2} \end{aligned}$$

So $d_{2j} \lesssim 2^{-j}$ and $d_j \lesssim 2^{-j/2}$.

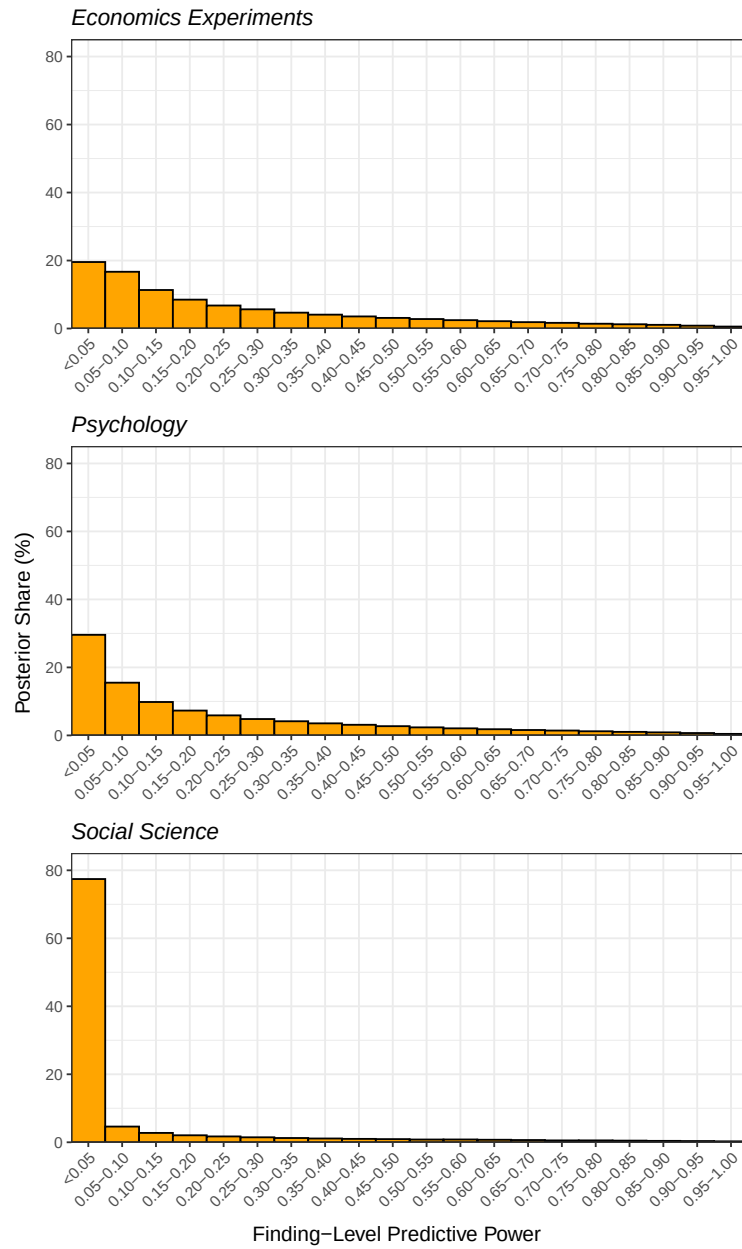
□

C. ADDITIONAL EMPIRICAL RESULTS

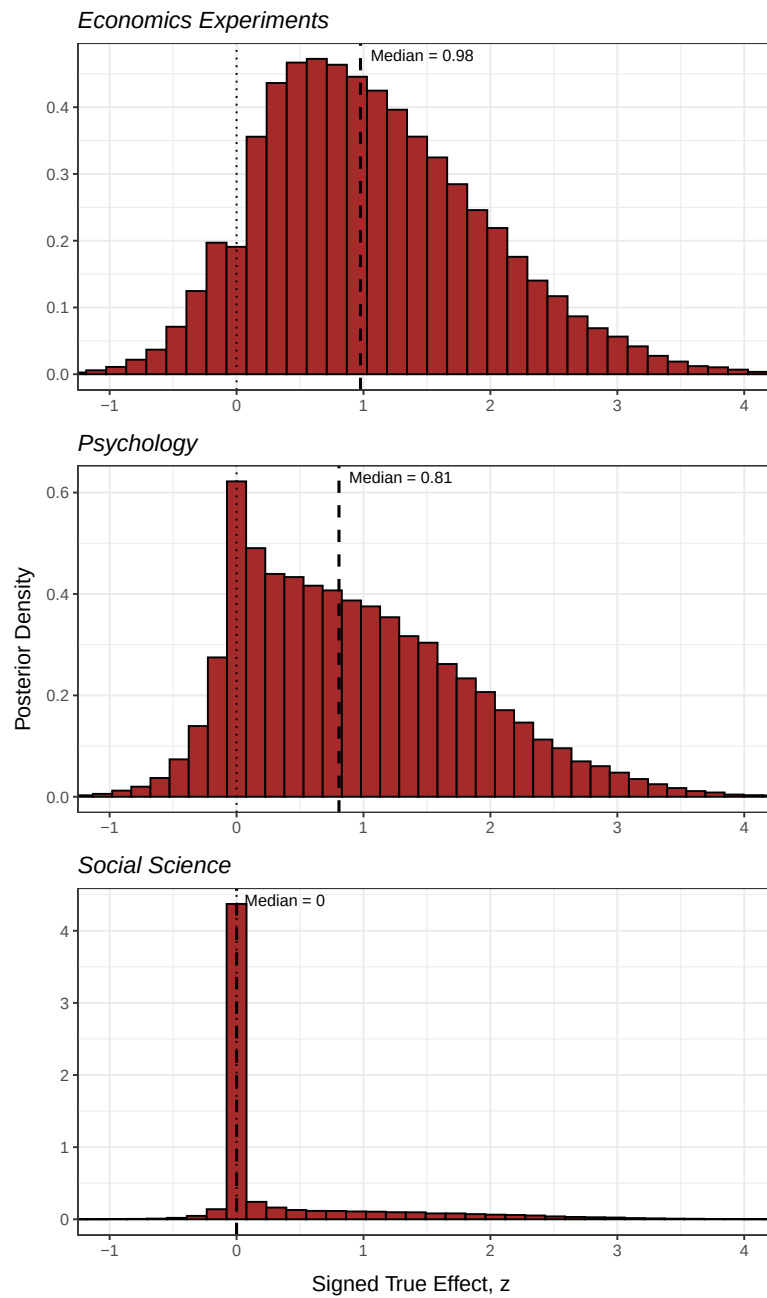
FIGURE C1. Predictive Power Curves, Systematic Replication Approach



Notes: The figure plots estimated predictive power curves for experimental economics, psychology, and experimental social science, using the systematic replication approach to estimate $\hat{\pi}(z)$. Parameter estimates for the distribution of true effects are taken from Table 1 (Economics), Table 2 (Psychology), and Table 11 (Social Science) in [Andrews and Kasy \(2019\)](#). All applications assume $\pi(\cdot)$ follows a gamma distribution. Predictive power is the probability that a replication with the same sample size as the original study produces a statistically significant estimate, conditional on the original study's two-sided p -value. Square brackets report the percentage of original studies with p -values at or below the indicated threshold. Underlying data are from [Camerer et al. \(2016\)](#), [Open Science Collaboration \(2015\)](#), and [Camerer et al. \(2018\)](#), respectively.

FIGURE C2. Distribution of Predictive Power Conditional on $p = 0.05$ 

Notes: The figure shows the distribution of replication probabilities conditional on an original two-sided p -value of 0.05. Each bar reports the share of findings whose probability of a successful replication falls within the indicated decile. A successful replication is statistically significant at the 5% level and has the same sign as the original estimate. Replications are assumed to use the same sample size as the original study. Estimates of the distribution of true effects are based on the metastudy replication approach. Underlying data are from [Camerer et al. \(2016\)](#), [Open Science Collaboration \(2015\)](#), and [Camerer et al. \(2018\)](#), respectively.

FIGURE C3. Distribution of z Conditional on $p = 0.05$ 

Notes: The figure plots the distribution of normalized true effects, z , conditional on an original two-sided p -value of 0.05. The vertical dotted line marks $z = 0$, and the vertical dashed line marks the median of the conditional distribution. Estimates of the distribution of true effects are based on the metastudy replication approach. Underlying data are from [Camerer et al. \(2016\)](#), [Open Science Collaboration \(2015\)](#), and [Camerer et al. \(2018\)](#), respectively.

D. HIGH-POWERED REPLICATIONS OVERSTATE REPLICABILITY

The systematic replication projects used rules to set sample sizes that generally led to larger samples than the original studies. The economics and psychology projects used a common-power rule that treats the original estimate as the true effect and targets a specified nominal power (Open Science Collaboration, 2015; Camerer et al., 2016). Because original estimates are noisy and tend to overstate true effects, expected power is below the nominal target; see Vu (2024). The social-science project used a fractional-power rule, targeting 90% power against an effect equal to three-quarters of the original estimate (Camerer et al., 2018).

Table D1 compares model-based expected replication rates under the implemented high-powered sample-size rules with those under the counterfactual assumption that replications use the same sample size as the original studies. More specifically, for each design, the expected replication rate is calculated as the average across studies, $\frac{1}{N} \sum_{i=1}^N r(p_i; \sigma_{ri})$, where $r(p_i; \sigma_{ri})$ is the expected replication probability for study i given its original p -value and the replication standard error implied by the relevant sample-size rule.

Column 3 reports expected replication rates under the implemented higher-powered designs, while Column 4 reports the counterfactual rates if replications used the original sample sizes. The higher-powered designs raise expected replication rates from 0.49 to 0.57 in economics and from 0.46 to 0.53 in psychology and social science. Column 5 shows that, at the study level, these designs increase replication probabilities by 30–36% on average.²⁴ Thus, systematic replication projects tend to overstate replicability relative to replications conducted at the precision of the original studies.

TABLE D1 – Expected Replication Rates With Alternative Sample Size Setting Rules

Field	Rule	Implemented Design	Existing Design	Average Ratio
Economics	Common Power Rule (92%)	0.57	0.49	1.33
Psychology	Common Power Rule (92%)	0.53	0.46	1.30
Social science	Fractional Power Rule (0.75, 90%)	0.53	0.46	1.36

Notes: The table reports model-based expected replication rates under the implemented replication sample-size rules and under the counterfactual assumption that replications use the same sample size as the original studies. The final column reports the average across studies of the ratio of the implemented-design replication probability to the same-sample-size replication probability.

²⁴The final column averages the study-level ratio of high-powered to same-powered replication probabilities, rather than taking the ratio of the averages in Columns 3 and 4.

E. PUBLICATION BIAS ESTIMATES

Table E2 reports the estimated publication probability ratios corresponding to the estimates in Figure 4. Recall the definition of θ_m from Equation 8, which is the probability of publication conditional on $|\hat{z}|$ lying between the m th and $(m - 1)$ th critical values. Each row reports the estimated ratio of publication probabilities immediately above and below a given critical value. For example, the entry in the first row of the RCT column means that crossing the critical value of 1.64 increases publication probability by about 8%.

TABLE E2 – Estimated Publication-Bias Probability Ratios by Research Design

Critical Threshold	RCT	RDD	DID	IV
1.64	1.08	0.98	1.04	1.28
1.96	1.03	0.97	1.02	1.27
2.58	0.99	0.70	0.93	0.96